Uniform and Relative Non-Associativity on Minimal Free Resolutions

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Abstract

We study unital chain-map multiplications on minimal free resolutions of quotient rings. We prove that Katthän's example admits no associative multiplication over any field, without assuming graded commutativity or compatibility with an internal grading. The proof gives a relative obstruction in which associativity on a specified finite collection of triples forces failure at another triple. The argument extends to a family that includes nonmonomial ideals. We also give criteria for uniform non-associativity, where a fixed triple fails for every multiplication, by comparing either associators or their primitives under changes of multiplication. These criteria persist from monomial elimination ideals and give explicit obstructions for Avramov's example and the known eight-vertex Bier sphere over every field. As an application, we show that the edge ideal of the eight-cycle admits an injective Taylor comparison on five selected generators, while every selection of that size fails to admit a multiplicative comparison. Four generators admit a multiplicative embedding.

Introduction

Let $(R, \mathfrak{m})$ be a local noetherian ring, and let $F \to R/I$ be a minimal free resolution with $F_0 = R$. A multiplication on F is a unital chain map $F \otimes_R F \to F$. We call it commutative when it is strictly graded-commutative, including the vanishing of odd squares. Buchsbaum and Eisenbud proved that a commutative multiplication always exists and asked whether an associative one can always be chosen [BE77]. Counterexamples were subsequently constructed in [Avr81, Sri92, Sri96, Kat19].

We study how associators change when the multiplication varies. Uniform non-associativity at a homogeneous triple means that its associator is nonzero for every multiplication. A relative obstruction imposes associativity on a prescribed collection of triples and forces its failure at another triple. Our main result uses this second form of obstruction to remove both the internal-grading and graded-commutativity hypotheses from the nonexistence theorem for Katthän's example.

Katthän proved that the minimal resolution of $\Bbbk[x,y,z,w]/\langle x^2, w^2, zw, xy, y^2z^2\rangle$ admits no commutative DG algebra structure whose multiplication respects the standard internal grading [Kat19, Theorem 5.1]. We prove the following statement for all unital chain-map multiplications.

Theorem A (Unrestricted relative non-associativity)**.** Over every field $\Bbbk$, the minimal resolution over $\Bbbk[x,y,z,w]$ of

$$\Bbbk[x,y,z,w]/\langle x^2, w^2, zw, xy, y^2z^2\rangle$$

admits no associative multiplication. Associativity on the 32 core triples specified in (28) forces non-associativity at (e_1, e_5, e_2). The same assertions hold after localization at $\langle x,y,z,w\rangle$ and completion there.

Theorems 3.2 and 3.3 prove this statement. Hancharuk, Laurent-Gengoux, and Strobl compute an explicit A_∞-structure for the same ideal, including the reference associator primitive with coefficient yz [HLS24, Example 4.3]. Our contribution is the obstruction to simultaneous cancellation of the associators under arbitrary changes of multiplication. The familiar reference associator at (e_1, e_5, e_2) can be cancelled by a change of multiplication, as Example 3.1 shows. Associativity on the core triples constrains the same correction coefficients. Reduction modulo $\langle x, w\rangle$ and Nakayama's lemma then force a nonzero associator at the distinguished triple. Theorem 3.7 extends this argument by replacing y^2z^2 with $yz\,c(y,z)$ for any nonzero $c \in \langle y, z\rangle$, including nonhomogeneous polynomials.

The uniform criteria compare a reference associator with all of its homotopy corrections. The first criterion detects a class modulo $\mathfrak{m}^2F$ and endpoint products with boundaries.

Theorem B (Uniform non-associativity)**.** Let $(R, \mathfrak{m})$ be local and noetherian, and let F be a minimal free resolution of R/I with reference multiplication μ. If homogeneous $a_1, a_2, a_3 \in F$ satisfy

$$\mathrm{d}a_i \in \mathfrak{m}^2F, \qquad a_1a_2, a_2a_3 \in \mathfrak{m}F, \qquad [a_1, a_2, a_3]_\mu \notin \mathfrak{m}^2F + a_1\mathrm{d}F + a_3\mathrm{d}F,$$

then every multiplication on F is non-associative at this triple. For monomial ideals, these hypotheses persist from an elimination ideal to the original ideal when the reference multiplication is multigraded.

This combines Theorem 1.1 with Proposition 2.2. For degree-one inputs, Theorem 1.5 gives a stronger criterion by testing a primitive of the associator one homological degree higher. The primitive is compared modulo coefficient ideals, boundaries, and endpoint products. Proposition 1.8 gives coefficient-ideal estimates in arbitrary homological degrees. Powers of Avramov's example yield uniform obstructions with arbitrarily high coefficient order.

The elimination argument applies these criteria to the known eight-vertex Bier sphere, identifying an explicit fixed triple at which every multiplication fails over every field. Denham and Suciu studied this sphere and its Massey-product obstruction in the characteristic-zero setting of [DS07, Section 8], within Baskakov's family [Bas03]. Earlier grade-five Gorenstein counterexamples appear in [Sri96]. The permanence argument uses the retained-resolution construction of [Boo12] and is related to DG-sensitive pruning [GMP26].

Uniformity also obstructs multiplicative Taylor comparisons when both the target multiplication and the comparison map vary. For the edge ideal of the eight-cycle, an injective Taylor comparison on five selected generators exists, but every selection of that size fails to admit a multiplicative comparison. Four generators admit a multiplicative embedding. Section 4 proves this sharp result and a family of related cycle examples.

Section 1 develops the criteria and their first examples. Section 2 proves permanence under elimination and gives the path and Gorenstein applications. Section 3 proves the relative obstruction and its family extension, and Section 4 treats Taylor comparisons.

Throughout, degrees of maps and graded commutativity refer to the homological grading. For polynomial rings, $\mathfrak{m}$ denotes the homogeneous maximal ideal. Competing multiplications and homotopies may have arbitrary internal degrees unless stated otherwise.

1 Criteria for uniform non-associativity

Throughout this section, let R be a commutative ring, let $I \subsetneq R$ be an ideal, and let $F \to R/I$ be a nonnegative free resolution over R with $F_0 = R$.

1.1 Multiplications and generalized associators

Recall that by a multiplication on F we mean a unital chain map $\mu\colon F \otimes_R F \to F$, not necessarily associative. We call it a commutative multiplication when it is also strictly graded-commutative. Every multiplication on F lifts the usual multiplication on R/I: R-bilinearity and unitality give $\mu(r,s) = rs\mu(1,1) = rs$ for $r,s \in R$, and in positive degrees the augmentation is zero. For an arbitrary chain map $\alpha\colon F \otimes_R F \to F$, the lifting condition is instead $\alpha(1,1) - 1 \in I$; it is not automatic.

In particular, after fixing a multiplication μ, every other multiplication on F has the form $\mu_h := \mu + \mathrm{d}h + h\mathrm{d}$, where $h\colon F \otimes_R F \to F$ is a homotopy, that is, a graded R-linear map of homological degree one. This is the usual comparison-homotopy argument: $F \otimes_R F$ is a bounded-below complex of free modules, and the two maps have the same composite with the augmentation $F \to R/I$. No normalization of h is needed for the first criterion below.

For minimal free resolutions over local noetherian rings, Buchsbaum and Eisenbud showed in [BE77] that a commutative multiplication always exists. They asked whether an associative commutative multiplication can be chosen. It was subsequently shown that this need not be the case; see [Avr81, Kat19, Sri92, Sri96] for counterexamples. In this section, we show that if a multiplication μ fails to be associative at a homogeneous triple (a_1, a_2, a_3) and some additional conditions hold for μ and that triple, then every multiplication on F fails to be associative at the same triple.

Before we state and prove the theorem, we set up notation. Equip F with a fixed multiplication μ, and write $a_1a_2 = \mu(a_1, a_2)$. The associator for μ is the chain map $[\cdot]_\mu\colon F \otimes_R F \otimes_R F \to F$ given by $[\cdot]_\mu = \mu(\mu \otimes 1 - 1 \otimes \mu)$. The corresponding R-trilinear map is denoted $[\cdot,\cdot,\cdot]_\mu\colon F^3 \to F$. In particular, for $a_1, a_2, a_3 \in F$,

$$[a_1 \otimes a_2 \otimes a_3]_\mu = [a_1, a_2, a_3]_\mu = (a_1a_2)a_3 - a_1(a_2a_3).$$

Since μ is fixed, we simplify notation by dropping its subscript when there is no ambiguity. If $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$ is another multiplication, write $a \star b = \mu_h(a,b)$. Its associator is given by

$$[\cdot]_{\mu_h} = [\cdot] + \mathrm{d}H + H\mathrm{d}, \tag{1}$$

where $H = [\cdot]_{\mu,h} + [\cdot]_{h,\mu_h}$ and

$$[\cdot]_{\mu,h} = \mu(h \otimes 1 - 1 \otimes h), \qquad [\cdot]_{h,\mu_h} = h(\mu_h \otimes 1 - 1 \otimes \mu_h). \tag{2}$$

We refer to these maps as generalized associators because they generalize the usual associator in the sense that $[\cdot]_{\mu,\mu} = [\cdot]_\mu$. Additional signs appear when evaluating (2) on elements, according to the Koszul sign rule. In (1), we use the same symbol d for the differentials on F and $F^{\otimes 3}$, with the domain making its meaning clear; for instance, the differential in $\mathrm{d}H$ is that of F because H lands in F. We use the same convention for the differential on $F^{\otimes 2}$.

Finally, we can decompose the generalized associator $[\cdot]_{h,\mu_h}$ into three generalized associators:

$$[\cdot]_{h,\mu_h} = [\cdot]_{h,\mu} + [\cdot]_{h,\mathrm{d}h} + [\cdot]_{h,h\mathrm{d}}.$$

The notation indicates how these maps are defined; for example, $[\cdot]_{h,h\mathrm{d}} = h(h\mathrm{d} \otimes 1 - 1 \otimes h\mathrm{d})$. With this notation in place, we are ready to state and prove the criterion.

1.2 The uniform criterion

Theorem 1.1 (Uniform non-associativity)**.** *Let $(R, \mathfrak{m})$ be a local noetherian ring and let F be a minimal free resolution of R/I over R, equipped with a multiplication μ. Suppose there are homogeneous elements $a_1, a_2, a_3 \in F$ such that*

$$\mathrm{d}a_1, \mathrm{d}a_2, \mathrm{d}a_3 \in \mathfrak{m}^2 F, \qquad a_1a_2, a_2a_3 \in \mathfrak{m}F,$$

and

$$[a_1, a_2, a_3] \notin \mathfrak{m}^2 F + a_1\mathrm{d}F + a_3\mathrm{d}F.$$

Then every multiplication on F fails to be associative at (a_1, a_2, a_3).

Proof. Let $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$ be any competing multiplication, by the comparison-homotopy argument above. We analyze the correction $\mathrm{d}H + H\mathrm{d}$ in (1) modulo $\mathfrak{m}^2F$. Set $a_{1,2} = h(a_1, a_2)$ and $a_{2,3} = h(a_2, a_3)$. We have

$$\begin{aligned}
(\mathrm{d}H + H\mathrm{d})(a_1 \otimes a_2 \otimes a_3) &\equiv \mathrm{d}H(a_1 \otimes a_2 \otimes a_3) \\
&= \mathrm{d}\big([\cdot]_{\mu,h} + [\cdot]_{h,\mu} + [\cdot]_{h,\mathrm{d}h} + [\cdot]_{h,h\mathrm{d}}\big)(a_1 \otimes a_2 \otimes a_3) \\
&\equiv \mathrm{d}\big([\cdot]_{\mu,h} + [\cdot]_{h,\mu}\big)(a_1 \otimes a_2 \otimes a_3) \\
&\equiv \mathrm{d}[\cdot]_{\mu,h}(a_1 \otimes a_2 \otimes a_3) \\
&= \mathrm{d}\big(a_{1,2}a_3 - (-1)^{|a_1|}a_1a_{2,3}\big) \\
&\equiv (\mathrm{d}a_{1,2})a_3 - a_1\mathrm{d}a_{2,3}.
\end{aligned}$$

The first congruence uses $\mathrm{d}a_i \in \mathfrak{m}^2F$. For the second, minimality gives $\mathrm{d}F \subseteq \mathfrak{m}F$, and the differential occurs twice in each of the discarded terms. The third uses $a_1a_2, a_2a_3 \in \mathfrak{m}F$, followed by the outer differential. The last congruence follows from the Leibniz rule and $\mathrm{d}a_1, \mathrm{d}a_3 \in \mathfrak{m}^2F$.

Consequently,

$$[a_1, a_2, a_3]_{\mu_h} \equiv [a_1, a_2, a_3] + (\mathrm{d}a_{1,2})a_3 - a_1\mathrm{d}a_{2,3} \pmod{\mathfrak{m}^2F}. \tag{3}$$

Let μ^{op} be the graded opposite multiplication, defined by

$$\mu^{\mathrm{op}}(a, b) = (-1)^{|a||b|}\mu(b, a).$$

The graded twist is a chain map, so μ^{op} is again a multiplication. The comparison-homotopy argument and minimality give the first inclusion below; R-bilinearity and $\mathrm{d}F \subseteq \mathfrak{m}F$ give the second:

$$(\mu - \mu^{\mathrm{op}})(F, F) \subseteq \mathfrak{m}F, \qquad (\mu - \mu^{\mathrm{op}})(\mathrm{d}F, F) \subseteq \mathfrak{m}^2F. \tag{4}$$

Thus the correction in (3) lies in $\mathfrak{m}^2F + a_1\mathrm{d}F + a_3\mathrm{d}F$. The nonmembership hypothesis therefore gives $[a_1, a_2, a_3]_{\mu_h} \neq 0$. □

Remark 1.2. Neither the reference multiplication nor the competing multiplication is required to be commutative.

For a minimal graded resolution over a polynomial ring, the criterion can be applied after localization at the homogeneous maximal ideal. Every polynomial multiplication, whether internally graded or not, localizes to a multiplication there.

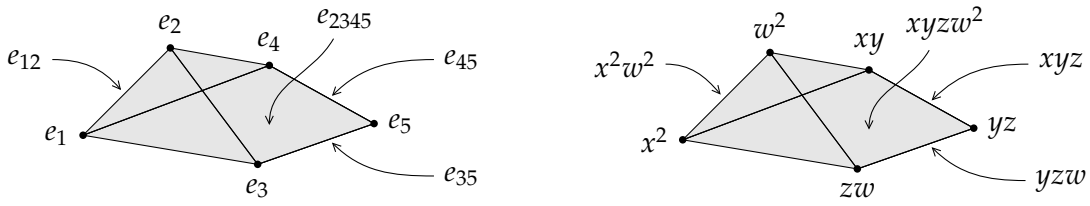
1.3 Avramov's example

We apply Theorem 1.1 to Avramov's example [Avr81, Example 2.2]. Avramov proves that its minimal resolution admits no DG module structure over the Koszul complex on the two squared generators. The Massey-product interpretation of this obstruction appears in [Avr81, Example 5.1.4], and the resulting nonexistence of a commutative DG algebra structure is stated explicitly in [Avr81, Remark 5.2.4]. We will show that one fixed triple is non-associative for every multiplication on the resolution.

Let $R = \Bbbk[x, y, z, w]$ be a polynomial ring over a field $\Bbbk$, let $\mathfrak{m} = \langle x, y, z, w \rangle$, and put

$$I = \langle x^2, w^2, zw, xy, yz \rangle.$$

Write F for the minimal free resolution of R/I over R. We use the conventions for labeled cellular resolutions of [BS98]; see also [MS05, Chapter 4]. One can visualize F as being supported on the I-labeled cellular complex below:



We write down the homogeneous components of F as a graded R-module:

$$\begin{aligned}
F_0 &= R, \\
F_1 &= Re_1 + Re_2 + Re_3 + Re_4 + Re_5, \\
F_2 &= Re_{12} + Re_{13} + Re_{14} + Re_{23} + Re_{24} + Re_{35} + Re_{45}, \\
F_3 &= Re_{123} + Re_{124} + Re_{1345} + Re_{2345}, \\
F_4 &= Re_{12345}.
\end{aligned}$$

The vertex labels, in order, are x^2, w^2, zw, xy, yz. We use compatible cellular orientations: on the simplicial faces the differential is the usual signed least-common-multiple differential, and, writing $\tau = e_{12345}$, we have

$$\mathrm{d}e_{1345} = x^2e_{35} - xwe_{45} - zwe_{14} + ye_{13}, \tag{5}$$

$$\mathrm{d}e_{2345} = xwe_{35} - w^2e_{45} - ze_{24} + xye_{23}, \tag{6}$$

$$\mathrm{d}\tau = -ye_{123} + ze_{124} - we_{1345} + xe_{2345}. \tag{7}$$

For example, $\mathrm{d}e_{123} = x^2e_{23} - we_{13} + ze_{12}$ and $\mathrm{d}e_{124} = xe_{24} - w^2e_{14} + ye_{12}$.

Choose a multigraded commutative multiplication μ on this resolution; such a choice is obtained by carrying out the usual lifting construction multidegree by multidegree. Multigrading and the Leibniz rule force the products needed below:

$$\begin{aligned}
e_1e_5 &= ze_{14} + xe_{45}, & e_2e_5 &= ye_{23} + we_{35}, \\
e_2e_{14} &= -e_{124}, & e_1e_{23} &= e_{123}, \\
e_2e_{45} &= -e_{2345}, & e_1e_{35} &= e_{1345}.
\end{aligned} \tag{8}$$

These formulas give

$$[e_1, e_5, e_2]_\mu = ye_{123} - ze_{124} + we_{1345} - xe_{2345} = -\mathrm{d}\tau. \tag{9}$$

The coefficients of this associator are linear. We verify that it satisfies all the hypotheses of Theorem 1.1.

Proposition 1.3. *Every multiplication $F \otimes_R F \to F$ is non-associative at (e_1, e_5, e_2). In particular, the conclusion holds for every commutative multiplication, without any assumption on its internal grading.*

Proof. All three vertex differentials x^2, yz, w^2 belong to $\mathfrak{m}^2$. By (8) and graded commutativity, both consecutive products e_1e_5 and e_5e_2 belong to $\mathfrak{m}F_2$. It remains to check the boundary-product terms in the criterion. The endpoint products with F_3 are

	e_{123}	e_{124}	e_{1345}	e_{2345}
$e_1\cdot$	0	0	0	$x\tau$
$e_2\cdot$	0	0	$-w\tau$	0.

Thus $e_1F_3, e_2F_3 \subseteq \mathfrak{m}F_4$. For $i \in \{1,2\}$ and $C \in F_3$, Leibniz gives

$$e_i\,\mathrm{d}C = (\mathrm{d}e_i)C - \mathrm{d}(e_iC) \in \mathfrak{m}^2F_3,$$

because $\mathrm{d}e_i \in \mathfrak{m}^2$ and the resolution is minimal. Consequently

$$\left(\mathfrak{m}^2F + e_1\mathrm{d}F + e_2\mathrm{d}F\right)_3 = \mathfrak{m}^2F_3.$$

The element $-\mathrm{d}\tau$ in (9) has nonzero linear coefficients, so it does not belong to this submodule. Apply Theorem 1.1 after localizing at $\mathfrak{m}$; the nonzero linear class remains nonzero, and any polynomial multiplication localizes.

Remark 1.2 shows that the competing multiplication need not be graded-commutative. In fact, the calculation above and (3) give the stronger congruence

$$[e_1, e_5, e_2]_\nu \equiv -\mathrm{d}\tau \not\equiv 0 \pmod{\mathfrak{m}^2F_3}.$$

□

1.4 A criterion using a primitive of the associator

Fix a reference multiplication $\mu\colon F \otimes_R F \to F$, write $ab = \mu(a,b)$, and write $a \star b = \mu_h(a,b)$ for a competing multiplication.

Theorem 1.1 detects an associator modulo $\mathfrak{m}^2F$ and endpoint products with boundaries. For degree-one triples, we can instead test a primitive one homological degree higher. Normalizing the comparison homotopy makes the terms involving the differentials of the three inputs vanish exactly, so no hypothesis $\mathrm{d}a_i \in \mathfrak{m}^2F$ is needed. The condition $\mathrm{d}F_3 \subseteq JF_2$ will control the boundaries of $h(a_1,a_2)$ and $h(a_2,a_3)$; for a minimal resolution and $J = \mathfrak{m}$, this condition is automatic. Both criteria establish non-associativity at one fixed triple for every multiplication.

Let $a_1, a_2, a_3 \in F$ be homogeneous, put $s = |a_1| + |a_2| + |a_3| > 0$, and suppose that $[a_1, a_2, a_3]$ is a cycle. Exactness gives $\tau \in F_{s+1}$ such that

$$\mathrm{d}\tau = [a_1, a_2, a_3]. \tag{10}$$

We call τ a *primitive of the associator*. Two choices differ by an element of $\mathrm{d}F_{s+2}$.

If $a_1, a_2, a_3 \in F_1$, the associator is automatically a cycle. Indeed, unitality and R-bilinearity put $R = F_0$ in the nucleus, meaning that every associator with a scalar input vanishes. The Leibniz rule for the associator then gives

$$\mathrm{d}[a_1, a_2, a_3] = [r_1, a_2, a_3] - [a_1, r_2, a_3] + [a_1, a_2, r_3] = 0,$$

where we set $r_i = \mathrm{d}a_i$.

Remark. The primitive formulation has a natural interpretation in terms of A_∞-structures. We use homological conventions: $\mu_n \colon F^{\otimes n} \to F$ has degree $n-2$, and the differential on the Hom complex is

$$\mathrm{d}^\star(\varphi) = \mathrm{d}\varphi - (-1)^{|\varphi|}\varphi\mathrm{d}.$$

With $\mu_2 = \mu$, we choose the sign convention for μ_3 so that the arity-three identity is

$$\mathrm{d}^\star(\mu_3) = \mathrm{d}\mu_3 + \mu_3\mathrm{d} = [\cdot]_\mu.$$

One can choose μ_3 to vanish whenever an input is 1, by the normalized lifting argument of Lemma 1.4. For $a_1, a_2, a_3 \in F_1$, this gives

$$\mathrm{d}\mu_3(a_1, a_2, a_3) = [a_1, a_2, a_3]_\mu.$$

Thus $\mu_3(a_1, a_2, a_3)$ is a choice of the primitive in (10). The existence and transfer of strictly unital A_∞-structures on free resolutions over R are standard; see [BCLP25, Section 5, especially Proposition 5.1].

Recall that for a reference chain map $\mu \colon F \otimes_R F \to F$ lifting the multiplication on R/I, every other such chain map has the form $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$ for some homotopy h. The lifting condition alone does not guarantee that h can be normalized. For multiplications, the two maps agree on scalar inputs, and the homotopy can be chosen to vanish there as well.

Lemma 1.4 (Normalized comparison)**.** *Let μ be a multiplication on F. Every other multiplication can be written as $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$, where $h \colon F \otimes_R F \to F$ is a graded R-linear map of homological degree one satisfying*

$$h(r, a) = h(a, r) = 0 \qquad (r \in R,\ a \in F).$$

If both μ and μ_h are commutative multiplications, then h can additionally be chosen to satisfy

$$h(a, b) = (-1)^{|a||b|}h(b, a), \qquad h(a, a) = 0 \quad \text{if } |a| \text{ is odd}$$

for homogeneous $a, b \in F$.

Proof. Put $\Delta = \mu_h - \mu$ and $U = F_0 \otimes_R F + F \otimes_R F_0$. The submodule U is a subcomplex of $F \otimes_R F$, and unitality implies that Δ vanishes on U. Thus Δ factors through a chain map $\overline{\Delta} \colon C = (F \otimes_R F)/U \to F$. The module C_n is the direct sum of $F_p \otimes_R F_q$ over $p, q > 0$ with $p + q = n$; in particular it is free, and $C_0 = C_1 = 0$.

Let $\epsilon \colon F \to R/I$ be the augmentation. Since $C_0 = 0$, we have $\epsilon\overline{\Delta} = 0$. The bounded-below complex C is degreewise free, so the homotopy lifting argument gives a graded map $\overline{h} \colon C \to F$ of degree one with

$$\overline{\Delta} = \mathrm{d}\overline{h} + \overline{h}\mathrm{d}_C.$$

Composing $\overline{h}$ with the quotient map $F \otimes_R F \to C$ gives the required normalized homotopy.

If both multiplications are commutative, let $V \subseteq F \otimes_R F$ be the graded submodule generated by

$$a \otimes b - (-1)^{|a||b|} b \otimes a, \qquad a \otimes a \quad \text{if } |a| \text{ is odd},$$

where a, b are homogeneous. It is a subcomplex: the symmetry relations are preserved by the differential, and for odd a, $\mathrm{d}(a \otimes a) = \mathrm{d}a \otimes a - a \otimes \mathrm{d}a$ is a symmetry relation. Now Δ factors through $D = (F \otimes_R F)/(U + V)$. A homogeneous basis of F containing 1 gives a basis of D indexed by unordered pairs of positive-degree basis elements, omitting repeated odd basis elements. Thus D is degreewise free and $D_0 = D_1 = 0$. The same homotopy lifting argument gives a homotopy vanishing on both U and V, proving the additional identities without division by two. □

Theorem 1.5 (Uniform non-associativity from a primitive). *Let $a_1, a_2, a_3 \in F_1$, and let $J \subseteq R$ be an ideal such that*

$$a_1a_2, a_2a_3 \in JF_2, \qquad \mathrm{d}F_3 \subseteq JF_2.$$

Set

$$N = JF_4 + \mathrm{d}F_5 + F_3a_3 + a_1F_3 \subseteq F_4.$$

For every multiplication μ_h and every primitive τ_h of its associator at (a_1, a_2, a_3), one has

$$\tau_h \equiv \tau \pmod{N}.$$

In particular, if $\tau \notin N$, then every multiplication on F is non-associative at (a_1, a_2, a_3). No hypothesis on the orders of $\mathrm{d}a_i$ is required.

Proof. Let μ_h be a competing multiplication, and choose a normalized homotopy h with $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$ by Lemma 1.4. Use the associator homotopy $H = [\cdot]_{\mu,h} + [\cdot]_{h,\mu_h}$ from (1). Since $\mathrm{d}a_i \in F_0 = R$, normalization and unitality give

$$h\mathrm{d}(a_i \otimes a_j) = 0, \qquad H\mathrm{d}(a_1 \otimes a_2 \otimes a_3) = 0.$$

Thus $\tau_h - \tau - H(a_1, a_2, a_3)$ is a cycle in F_4, hence belongs to $\mathrm{d}F_5 \subseteq N$. It remains to show that $H(a_1, a_2, a_3) \in N$.

Put $u = h(a_1, a_2)$ and $v = h(a_2, a_3)$, both in F_3. The pairwise vanishing above gives

$$a_1 \star a_2 = a_1a_2 + \mathrm{d}u \in JF_2, \qquad a_2 \star a_3 = a_2a_3 + \mathrm{d}v \in JF_2.$$

In generalized-associator notation, $[a_1, a_2, a_3]_{h,h\mathrm{d}} = 0$, so

$$\begin{aligned} H(a_1, a_2, a_3) = {} & [a_1, a_2, a_3]_{\mu,h} \\ & + [a_1, a_2, a_3]_{h,\mu} + [a_1, a_2, a_3]_{h,\mathrm{d}h}. \end{aligned} \tag{11}$$

Each term belongs to the corresponding summand of N:

$$\begin{aligned} [a_1, a_2, a_3]_{\mu,h} &= ua_3 + a_1v \in F_3a_3 + a_1F_3, \\ [a_1, a_2, a_3]_{h,\mu} &= h(a_1a_2, a_3) - h(a_1, a_2a_3) \in JF_4, \\ [a_1, a_2, a_3]_{h,\mathrm{d}h} &= h(\mathrm{d}u, a_3) - h(a_1, \mathrm{d}v) \in JF_4. \end{aligned}$$

The last two inclusions use the hypotheses and R-linearity of h, with $h(F_2 \otimes_R F_1), h(F_1 \otimes_R F_2) \subseteq F_4$. This proves the congruence. If μ_h were associative at the triple, we could choose $\tau_h = 0$, contradicting $\tau \notin N$. □

Corollary 1.6 (The residue-field criterion). *Suppose $(R, \mathfrak{m}, \Bbbk)$ is local and F is minimal. Put $V = F/\mathfrak{m}F$, and use a bar for reduction modulo $\mathfrak{m}$. If $a_1, a_2, a_3 \in F_1$ satisfy*

$$a_1a_2, a_2a_3 \in \mathfrak{m}F_2, \qquad \bar{\tau} \neq 0 \quad \text{in} \quad \frac{V_4}{V_3\bar{a}_3 + \bar{a}_1V_3},$$

then every multiplication on F is non-associative at (a_1, a_2, a_3).

Proof. Apply Theorem 1.5 with $J = \mathfrak{m}$. Minimality gives $\mathrm{d}F_3 \subseteq \mathfrak{m}F_2$ and $\mathrm{d}F_5 \subseteq \mathfrak{m}F_4$. □

The product on V is independent of the chosen multiplication on F, since $\mathrm{d}h + h\mathrm{d}$ has image in $\mathfrak{m}F$. It is the usual product on $\mathrm{Tor}^R(R/I, \Bbbk)$. In particular, the quotient in Corollary 1.6 is independent of the reference map. In the general ideal formulation of Theorem 1.5, the submodule N is defined using the fixed reference multiplication ab. Its endpoint products are taken with respect to this multiplication throughout the comparison.

The endpoint-product quotient in Corollary 1.6 has the form of the familiar indeterminacy of a triple Massey product [DS07, Section 5.1]. The primitive theorem formulates the comparison over R, keeping track of coefficient ideals and endpoint submodules while fixing the triple and allowing every competing multiplication.

Remark 1.7. For degree-one triples, Corollary 1.6 recovers Theorem 1.1. Indeed, suppose that the hypotheses of that theorem hold. If

$$\tau \in \mathfrak{m}F_4 + F_3a_3 + a_1F_3,$$

then applying d, using minimality and $\mathrm{d}a_1, \mathrm{d}a_3 \in \mathfrak{m}^2F_0$, gives

$$[a_1, a_2, a_3]_\mu \in \mathfrak{m}^2F_3 + (\mathrm{d}F_3)a_3 + a_1\mathrm{d}F_3 \subseteq \mathfrak{m}^2F + a_1\mathrm{d}F + a_3\mathrm{d}F.$$

The last inclusion follows from (4), so this contradicts the nonmembership hypothesis. This also shows that the hypothesis $\mathrm{d}a_2 \in \mathfrak{m}^2F_0$ can be omitted from Theorem 1.1 when all three inputs have degree one. The original theorem continues to apply to triples of arbitrary homogeneous degrees. The converse fails even for degree-one triples: in Example 1.9, the displayed associator belongs to $\mathfrak{m}^2F_3$, while its primitive has nonzero residue in the quotient of Corollary 1.6. Thus the primitive criterion detects uniform non-associativity that Theorem 1.1 does not detect.

1.5 Controlling the correction by ideals

The next statement keeps track of the coefficient ideals used in the proof of Theorem 1.1.

Proposition 1.8 (An ideal-valued criterion)**.** *Let $a_1, a_2, a_3 \in F$ be homogeneous, and let $J, K, L \subseteq R$ be ideals satisfying*

$$\mathrm{d}F \subseteq JF, \qquad \mathrm{d}a_1, \mathrm{d}a_2, \mathrm{d}a_3 \in KF, \qquad a_1a_2, a_2a_3 \in LF.$$

For every multiplication μ_h on F,

$$[a_1, a_2, a_3]_{\mu_h} - [a_1, a_2, a_3]_\mu \in (K + JL + J^2)F + (\mathrm{d}F)a_3 + a_1\mathrm{d}F.$$

All products on the right are taken using μ. Consequently, if the reference associator lies outside this submodule, every such μ_h is non-associative at the same triple.

Proof. Let μ_h be a competing multiplication, and choose a homotopy h with $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$. Put $t = a_1 \otimes a_2 \otimes a_3$. By (1),

$$[t]_{\mu_h} - [t]_\mu = (\mathrm{d}H + H\mathrm{d})(t),$$

where

$$H = [\cdot]_{\mu,h} + [\cdot]_{h,\mu_h} = [\cdot]_{\mu,h} + [\cdot]_{h,\mu} + [\cdot]_{h,\mathrm{d}h} + [\cdot]_{h,h\mathrm{d}}.$$

The hypotheses give

$$H\mathrm{d}(t) \in KF, \qquad \mathrm{d}[t]_{h,\mu} \in JLF, \qquad \mathrm{d}[t]_{h,\mathrm{d}h},\ \mathrm{d}[t]_{h,h\mathrm{d}} \in J^2F.$$

The first inclusion uses $\mathrm{d}a_i \in KF$. For the second, the adjacent products lie in LF, and applying d contributes a factor of J. For the last two, both $\mathrm{d}h$ and $h\mathrm{d}$ have image in JF; applying the outer h preserves this coefficient ideal, and applying d contributes another factor of J.

It remains to consider $\mathrm{d}[t]_{\mu,h}$. Set $u = h(a_1, a_2)$ and $v = h(a_2, a_3)$. The Koszul sign rule and the Leibniz identity give

$$\begin{aligned}\mathrm{d}[t]_{\mu,h} &= \mathrm{d}\big(ua_3 - (-1)^{|a_1|}a_1 v\big)\\ &\equiv (\mathrm{d}u)a_3 - a_1\mathrm{d}v \pmod{KF}.\end{aligned}$$

The omitted terms contain $\mathrm{d}a_1$ or $\mathrm{d}a_3$, so they belong to KF. The remaining terms lie in $(\mathrm{d}F)a_3 + a_1\mathrm{d}F$. Combining these inclusions proves the assertion. □

For a minimal resolution over a local ring, taking $J = L = \mathfrak{m}$ and $K = \mathfrak{m}^2$ recovers Theorem 1.1, using (4) to move a_3 to the left modulo $\mathfrak{m}^2F$. More generally, for integers $s \geq 1$ and $t, u \geq 0$, if

$$\mathrm{d}F \subseteq \mathfrak{m}^s F, \qquad \mathrm{d}a_i \in \mathfrak{m}^t F, \qquad a_1a_2, a_2a_3 \in \mathfrak{m}^u F,$$

then the coefficient ideal in the proposition is $\mathfrak{m}^q$, where

$$q = \min\{t, s+u, 2s\}.$$

Thus the exponent two in the original criterion comes from minimality and the coefficient orders of the correction terms. The identity $\mathrm{d}^2 = 0$ does not make an expression such as $\mathrm{d}h\mathrm{d}$ vanish.

For degree-one triples, Theorem 1.5 also subsumes Proposition 1.8, using coefficient ideal $J + L$. Indeed, for

$$N' = (J+L)F_4 + \mathrm{d}F_5 + F_3a_3 + a_1F_3,$$

the Leibniz rule gives

$$\mathrm{d}N' \subseteq (K + JL + J^2)F_3 + (\mathrm{d}F_3)a_3 + a_1\mathrm{d}F_3.$$

Thus the associator nonmembership condition forces its primitive outside N'. The two associator criteria retain their scope in arbitrary homogeneous degrees.

Example 1.9 (Powers of Avramov's example)**.** Let $R = \Bbbk[x, y, z, w]$, let $\mathfrak{m} = \langle x, y, z, w\rangle$, and put

$$I_N = \langle x^{2N}, w^{2N}, z^N w^N, x^N y^N, y^N z^N\rangle, \qquad N \geq 2.$$

Apply the substitution $(x, y, z, w) \mapsto (x^N, y^N, z^N, w^N)$ to the resolution and reference multiplication in Subsection 1.3. The ring R is free over $\Bbbk[x^N, y^N, z^N, w^N]$, with basis the monomials whose exponents are less than N. Thus substitution preserves exactness. All differential entries remain in $\mathfrak{m}$, so the resulting resolution F is minimal. Retain the earlier basis labels and write $F_4 = Re_{12345}$.

The substituted associator is

$$[e_1, e_5, e_2]_\mu = y^N e_{123} - z^N e_{124} + w^N e_{1345} - x^N e_{2345} = -\mathrm{d}e_{12345}. \tag{12}$$

It lies in $\mathfrak{m}^N F_3$, so Theorem 1.1 does not detect it. Put $\mathfrak{q}_N = \langle x^N, y^N, z^N, w^N\rangle$. The substituted formulas give

$$\mathrm{d}F \subseteq \mathfrak{q}_N F, \qquad \mathrm{d}e_1, \mathrm{d}e_5, \mathrm{d}e_2 \in \mathfrak{q}_N^2, \qquad e_1e_5, e_5e_2 \in \mathfrak{q}_N F_2.$$

The only nonzero products of e_1 or e_2 with the basis of F_3 are

$$e_1e_{2345} = x^N e_{12345}, \qquad e_2e_{1345} = -w^N e_{12345}.$$

The Leibniz rule therefore puts the endpoint terms $(\mathrm{d}F_3)e_2 + e_1\mathrm{d}F_3$ in $\mathfrak{q}_N^2 F_3$. Applying Proposition 1.8 with $J = L = \mathfrak{q}_N$ and $K = \mathfrak{q}_N^2$ gives

$$[e_1, e_5, e_2]_{\mu_h} \equiv -\mathrm{d}e_{12345} \not\equiv 0 \pmod{\mathfrak{q}_N^2 F_3}$$

for every multiplication μ_h.

Corollary 1.6 gives the same conclusion after localization at $\mathfrak{m}$. A primitive is $\tau = -e_{12345}$. Its residue is nonzero, while the endpoint products vanish modulo $\mathfrak{m}$. For $N = 2$, this gives a quadratic associator detected by the primitive and ideal-valued criteria. In general, the reference associator has coefficient order exactly N. The conclusions hold over every field, and after localization or completion at $\mathfrak{m}$.

2 Permanence and examples

In this section, we consider monomial ideals in polynomial rings over a field $\Bbbk$ and their minimal free resolutions over the respective polynomial rings, equipped with the fine multigrading and with degree-zero module equal to the base ring.

2.1 Passing the criteria from an elimination ideal

The zero-new-variable part of the resolution below is the pruned resolution of Boocher [Boo12, Theorem 2.3]. We include the block-matrix argument to record the specialization retraction and the compatibility with the reference multiplication needed for our obstruction. Geller, Martin, and Potts-Rubin prove that commutative DG algebra structures descend under this pruning for squarefree monomial ideals [GMP26, Theorem 5.7 and Corollary 5.8]. Our permanence statement instead carries a specified fixed-triple obstruction from the smaller resolution to the larger one, and allows arbitrary multiplications as competitors.

Let $R = \Bbbk[x_1, \dots, x_n]$, let $S = R[y_1, \dots, y_m]$, and write

$$\mathfrak{m} = \langle x_1, \dots, x_n \rangle, \qquad \mathfrak{n} = \langle x_1, \dots, x_n, y_1, \dots, y_m \rangle.$$

Let $J \subsetneq S$ be a monomial ideal, and let $I = J \cap R$ be its *elimination ideal*, obtained by eliminating $y_1, \dots, y_m$. Equivalently, I is the contraction of J along $R \hookrightarrow S$. In this monomial setting, elimination also agrees with specialization to zero: if $\pi\colon S \to R$ sets $y_1 = \cdots = y_m = 0$, then $\pi(J) = I$. Indeed, each monomial generator either involves a new variable and vanishes, or belongs to R and survives unchanged. For squarefree ideals, the Stanley–Reisner complex of I is the induced subcomplex of the Stanley–Reisner complex of J on the original vertices.

Lemma 2.1. *Let $J \subsetneq S$ be a monomial ideal, put $I = J \cap R$, and let F and G be the minimal free resolutions of R/I over R and of S/J over S, respectively. After a multigraded identification, $F \otimes_R S$ is a subcomplex of G, and*

$$G \otimes_S R \cong F \oplus Q$$

as R-complexes, where $S \to R$ sets $y_1, \dots, y_m$ equal to zero. Every multigraded multiplication μ on F extends to a multigraded multiplication μ' on G. If μ is commutative, the extension μ' can be chosen commutative. For this extension, there is an R-linear map $\rho\colon G \to F$ satisfying

$$\rho(sa) = s|_{y=0}\,\rho(a), \qquad \rho|_F = \mathrm{id}_F, \qquad \rho\mathrm{d} = \mathrm{d}\rho, \qquad \rho\mu'(a,b) = \mu(\rho(a), \rho(b)).$$

Proof. Choose a multihomogeneous basis of G. Its basis degrees have nonnegative coordinates. Let E_i be the S-span of the basis elements of G_i whose degrees in all the new variables are zero, and let P_i be the S-span of the remaining basis elements. Thus $G_i = E_i \oplus P_i$ as free S-modules. A differential term from E_i to P_{i-1} would require a coefficient with a negative degree in some new variable, which is impossible. Hence E is a subcomplex, and its differential has coefficients in R.

Taking the part of new-variable degree zero in the augmented resolution $G \to S/J$ over S preserves exactness. This part is the R-span of the basis elements defining E, and its zeroth homology is R/I, since $J \cap R = I$. It is a minimal free resolution over R, which we identify with F. Consequently $E = F \otimes_R S$.

To see the splitting after specialization explicitly, write the differential with respect to $G_i = E_i \oplus P_i$ as

$$\mathrm{d}_i^G = \begin{pmatrix} \mathrm{d}_i^E & A_i \\ 0 & B_i \end{pmatrix}.$$

A term from a basis element of P_i to one of E_{i-1} has a coefficient of positive degree in at least one new variable. Thus every entry of A_i belongs to $\langle y_1, \dots, y_m \rangle$ and vanishes after setting $y = 0$. Since $E \otimes_S R \cong F$, the specialized differential is

$$\mathrm{d}_i^{G \otimes_S R} = \begin{pmatrix} \mathrm{d}_i^F & 0 \\ 0 & \overline{B}_i \end{pmatrix}, \qquad \overline{B}_i = B_i|_{y=0}.$$

In particular, the modules $Q_i = P_i \otimes_S R$ with differentials $\overline{B}_i$ form an R-complex Q, and the displayed block diagonal differential gives

$$G \otimes_S R \cong F \oplus Q$$

as R-complexes. The modules P_i need not form a subcomplex of G before specialization. Define ρ as the composite

$$G \longrightarrow G \otimes_S R \cong F \oplus Q \longrightarrow F,$$

where the first map specializes coefficients and the last map is the projection. This proves the first three identities.

To extend an arbitrary multiplication, take $C = G \otimes_S G$. If μ is commutative and we want a commutative extension, instead take the quotient complex

$$C = (G \otimes_S G)/\langle u \otimes v - (-1)^{|u||v|} v \otimes u,\ u \otimes u \text{ for } |u| \text{ odd} \rangle,$$

where the relations are taken on homogeneous elements. The differential descends to this quotient. In particular, the differential of an odd square is $\mathrm{d}u \otimes u - u \otimes \mathrm{d}u$, which is zero modulo the commutativity relations. The complex C is degreewise free on ordered basis pairs in the first case, and on unordered basis pairs, omitting repeated odd basis elements, in the second. In either case, the unit pairs and the pairs from E span a subcomplex which is a direct summand in each degree. Prescribe the product there using the unit and $\mu \otimes_R S$. These prescriptions agree on their intersection.

Extend this chain map inductively to the remaining basis pairs. Each such pair has total homological degree at least two. At each stage its required differential is a cycle in a positive homological degree of G, so exactness supplies a lift. We may choose the lift in the required multidegree. The resulting map $C \to G$ gives a multiplication μ' extending μ, commutative in the second case. This construction involves no division by two.

Finally, nonnegativity of the new-variable degrees shows that a product with an input of positive new-variable degree also has positive new-variable degree. The map ρ kills all such elements. On inputs of zero new-variable degree the product is μ. This gives the last identity. □

The splitting in Lemma 2.1 is a splitting after specialization. In general $F \otimes_R S$ is not a direct summand of G as an S-complex. For example, for $I = \langle x \rangle \subseteq \Bbbk[x]$ and $J = \langle x, y \rangle \subseteq \Bbbk[x, y]$, a chain retraction from the Koszul resolution over $\Bbbk[x, y]$ would have to send the generator with differential y to ae_x with $ax = y$, which is impossible.

Proposition 2.2 (Persistence from an elimination ideal)**.** *With the notation of Lemma 2.1, suppose a multigraded multiplication μ on F and homologically homogeneous elements $a_1, a_2, a_3 \in F$ satisfy*

$$\mathrm{d}a_i \in \mathfrak{m}^2 F \quad (1 \le i \le 3), \qquad a_1 a_2, a_2 a_3 \in \mathfrak{m}F,$$

and

$$[a_1, a_2, a_3]_\mu \notin \mathfrak{m}^2 F + a_1 \mathrm{d}F + a_3 \mathrm{d}F.$$

Then every multiplication on G is non-associative at the same triple (a_1, a_2, a_3), viewed in G. The competing multiplication need not respect the internal grading or be graded-commutative.

Proof. Choose μ' and ρ as in Lemma 2.1. The differential and the reference multiplication on the embedded F are unchanged, so the first two hypotheses persist with $\mathfrak{n}$ in place of $\mathfrak{m}$. Moreover,

$$\rho(\mathfrak{n}^2 G) \subseteq \mathfrak{m}^2 F, \qquad \rho(a_i \mathrm{d}G) \subseteq a_i \mathrm{d}F \quad (i = 1, 3).$$

If the associator belonged to $\mathfrak{n}^2 G + a_1 \mathrm{d}G + a_3 \mathrm{d}G$, applying ρ would contradict the assumed nonmembership in F. Thus all three hypotheses of the uniform criterion hold for μ'.

The quotient by $\mathfrak{n}^2 G + a_1 \mathrm{d}G + a_3 \mathrm{d}G$ is annihilated by $\mathfrak{n}^2$, so this nonzero class remains nonzero after localization at $\mathfrak{n}$. Apply Theorem 1.1 and Remark 1.2. Every polynomial multiplication localizes, which proves the assertion. □

The same specialization map preserves the residue-field primitive obstruction of Theorem 1.5; we record this stronger permanence statement in Corollary 2.3.

Corollary 2.3 (Persistence of the primitive obstruction)**.** *Use the notation of Lemma 2.1, and write $\mathfrak{m} = \langle x_1, \dots, x_n \rangle$ and $\mathfrak{n} = \langle x_1, \dots, x_n, y_1, \dots, y_m \rangle$. Let μ be a multigraded multiplication on F, let $a_1, a_2, a_3 \in F_1$, and suppose*

$$a_1 a_2, a_2 a_3 \in \mathfrak{m}F_2, \qquad \tau_\mu \notin \mathfrak{m}F_4 + \mathrm{d}F_5 + a_3 F_3 + a_1 F_3,$$

where $\mathrm{d}\tau_\mu = [a_1, a_2, a_3]_\mu$. After identifying F with its old-variable part in G, every multiplication on G is non-associative at (a_1, a_2, a_3).

Proof. Choose the extended reference multiplication μ' and the specialization map ρ from Lemma 2.1. The element τ_μ is still a primitive of the reference associator in G, because both the differential and the reference product restrict to those on F. By (4), the submodules $F_3 a_3 + \mathfrak{m}F_4$ and $a_3 F_3 + \mathfrak{m}F_4$ coincide; the same holds for G and $\mathfrak{n}$. Minimality gives $\mathrm{d}G_3 \subseteq \mathfrak{n}G_2$, and the two adjacent products belong to $\mathfrak{n}G_2$. Moreover,

$$\rho\big(\mathfrak{n}G_4 + \mathrm{d}G_5 + a_3 G_3 + a_1 G_3\big) \subseteq \mathfrak{m}F_4 + \mathrm{d}F_5 + a_3 F_3 + a_1 F_3.$$

Here we use semilinearity, $\rho\mathrm{d} = \mathrm{d}\rho$, multiplicativity of ρ for the reference products, and $\rho(a_i) = a_i$. Since $\rho(\tau_\mu) = \tau_\mu$, the assumed nonmembership in F implies the corresponding nonmembership in G. Theorem 1.5, with coefficient ideal $\mathfrak{n}$, proves the result. No restriction of an arbitrary competing product on G to F is used. □

2.2 The path on six vertices

Let

$$R = \Bbbk[x_1, \dots, x_6], \qquad I = \langle x_1x_2, x_2x_3, x_3x_4, x_4x_5, x_5x_6 \rangle.$$

This is the edge ideal of the path on six vertices, and it is the polarization of Avramov's ideal. Its nonexistence of a graded-commutative DG algebra structure is also discussed in [Kat19, Example 3.8]. We record the calculation needed to apply our uniform criterion.

Use the same cellular basis as in Subsection 1.3, now denoted by η_σ, with vertex labels ordered as

$$\mathrm{d}\eta_1 = x_1x_2, \quad \mathrm{d}\eta_2 = x_5x_6, \quad \mathrm{d}\eta_3 = x_4x_5, \quad \mathrm{d}\eta_4 = x_2x_3, \quad \mathrm{d}\eta_5 = x_3x_4.$$

The labels are obtained by polarizing x^2 to x_1x_2 and w^2 to x_5x_6, with x, y, z, w replaced by x_2, x_3, x_4, x_5, respectively. Polarization preserves the divisibility subcomplexes in the cellular acyclicity criterion, so the resulting labeled complex is a minimal resolution over R, with ranks $(1, 5, 7, 4, 1)$. Set $\tau_{\mathrm{P}} = \eta_{12345}$. The top differential is

$$\mathrm{d}\tau_{\mathrm{P}} = -x_3\eta_{123} + x_4\eta_{124} - x_6\eta_{1345} + x_1\eta_{2345}. \tag{13}$$

Choose a multigraded commutative multiplication μ_{P}. As before, multigrading and the Leibniz rule force

$$\begin{aligned} \eta_1\eta_5 &= x_4\eta_{14} + x_1\eta_{45}, & \eta_2\eta_5 &= x_3\eta_{23} + x_6\eta_{35}, \\ \eta_2\eta_{14} &= -\eta_{124}, & \eta_1\eta_{23} &= \eta_{123}, \\ \eta_2\eta_{45} &= -\eta_{2345}, & \eta_1\eta_{35} &= \eta_{1345}. \end{aligned}$$

Hence

$$[\eta_1, \eta_5, \eta_2]_{\mu_{\mathrm{P}}} = -\mathrm{d}\tau_{\mathrm{P}}.$$

The endpoint products with F_3 vanish except for

$$\eta_1\eta_{2345} = x_2\tau_{\mathrm{P}}, \qquad \eta_2\eta_{1345} = -x_5\tau_{\mathrm{P}}.$$

Since $\mathrm{d}\eta_1, \mathrm{d}\eta_2 \in \mathfrak{m}^2$, the Leibniz rule gives

$$\eta_1\mathrm{d}F_3 + \eta_2\mathrm{d}F_3 \subseteq \mathfrak{m}^2F_3.$$

Thus the nonzero linear associator in (13) satisfies the nonmembership condition. The other two hypotheses follow from the quadratic vertex differentials and the displayed products. Theorem 1.1 and Remark 1.2 therefore give uniform non-associativity at (η_1, η_5, η_2) for every multiplication.

2.3 A Gorenstein example

Denham and Suciu identify $\mathrm{Bier}(C_4)$ with their eight-vertex sphere and exhibit a nonzero triple Massey product in the characteristic-zero setting of [DS07, Section 8, especially Examples 8.1.1 and 8.6.1]. Their Remark 8.1.2 places the associated moment-angle manifold in Baskakov's family [Bas03]. Grade-five Gorenstein counterexamples were previously constructed in [Sri96]. The theorem below identifies an explicit fixed triple at which every multiplication fails over every field.

More explicitly, take C_4 on vertices $1,2,3,4$ with edges $12, 23, 34, 41$, and number the vertices of its Alexander dual by $i' = i + 4$. The vertex map

$$(1,2,3,4,5,6,7,8) \longmapsto (1,3,7,6,2,4,8,5)$$

identifies the Stanley–Reisner complex of (14) with the deleted join $\mathrm{Bier}(C_4) = C_4 *_\Delta C_4^\vee$.

Theorem 2.4. *Let $S = \Bbbk[x_1, \dots, x_8]$ and let*

$$\begin{aligned} J = \langle\, & x_1x_2, x_2x_3, x_3x_4, x_4x_5, x_5x_6, \\ & x_3x_7, x_6x_7, x_1x_8, x_4x_8, x_7x_8 \rangle. \end{aligned} \tag{14}$$

Then S/J is Gorenstein of dimension 3 and codimension 5 over every field. Let G be its minimal multigraded free resolution over S, and let $b_i \in G_1$ be the basis element with $\mathrm{d}b_i = x_ix_{i+1}$ for $1 \le i \le 5$. Every multiplication on G is non-associative at (b_1, b_3, b_5). In particular, G admits no DG algebra structure, even when compatibility with the internal grading and graded commutativity are not required.

Proof. We first verify the Gorenstein assertion. Write ijk for the face $\{i, j, k\}$. The Stanley–Reisner complex of the path ideal I has facets

$$135, \quad 136, \quad 146, \quad 246, \quad 25.$$

Adjoining x_7 and the generators x_3x_7, x_6x_7 gives the complex B with facets

$$135, \quad 136, \quad 146, \quad 246, \quad 147, \quad 157, \quad 247, \quad 257.$$

This is a triangulated disk. Indeed, in the indicated order each triangle after the first is attached along one boundary edge or two consecutive boundary edges; the successive attaching intersections are

$$13, \quad 16, \quad 46, \quad 14, \quad 15 \cup 17, \quad 24 \cup 47, \quad 27 \cup 57.$$

The boundary of B is the cycle $2 - 5 - 3 - 6 - 2$. Adjoining x_8 and the generators x_1x_8, x_4x_8, x_7x_8 adds exactly the four triangles

$$258, \quad 358, \quad 368, \quad 268.$$

Thus the Stanley–Reisner complex of J is $B \cup (8 * \partial B)$, a triangulated 2-sphere. Its Stanley–Reisner ring is Gorenstein over every field [Sta96, Theorem II.5.1]. Since the facets have three vertices and S has eight variables, its dimension is 3 and its codimension is 5.

Finally, $J \cap \Bbbk[x_1, \dots, x_6] = I$. By Lemma 2.1, we may identify the path resolution with the part of G of zero degree in x_7, x_8. Under this identification, (η_1, η_5, η_2) becomes (b_1, b_3, b_5). The calculation in Subsection 2.2 and Proposition 2.2 prove the asserted uniform non-associativity. □

3 Katthän's example and relative non-associativity

Let $\Bbbk$ be a field, let $R = \Bbbk[x, y, z, w]$, let $\mathfrak{m} = \langle x, y, z, w \rangle$, let $I = \langle x^2, w^2, zw, xy, y^2z^2 \rangle$, and let F be the minimal multigraded free resolution of R/I over R, with $F_0 = R$. The ideal I is obtained from Avramov's ideal by replacing its last generator yz by y^2z^2.

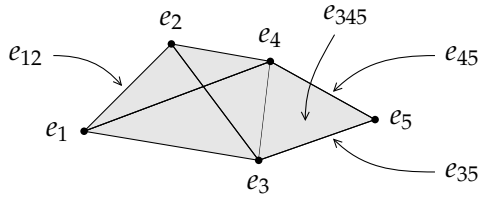
Katthän's theorem excludes graded-commutative DG algebra structures whose multiplication respects the standard internal grading [Kat19, Theorem 5.1]. Taking the degree-zero component of a multiplication preserves the chain-map and unit conditions, but need not preserve associativity, since components of opposite internal degrees also contribute to the degree-zero associativity equation. Removing the grading hypothesis therefore requires an additional argument. Moreover, Example 3.1 shows that even among standard-graded commutative multiplications, the associator at (e_1, e_5, e_2) can be cancelled. Our proof instead controls several triples simultaneously and shows that no multiplication on F is associative, even without graded commutativity.

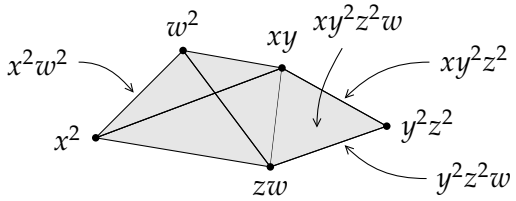
3.1 The resolution

In this subsection we describe F as a resolution over R. The ideal I is strongly generic: no variable appears with the same nonzero exponent in two distinct minimal generators. Its Scarf complex is

$$\Delta = \langle \{1,2,3,4\}, \{3,4,5\} \rangle.$$

By [BPS98, Theorem 3.2], this complex supports the minimal free resolution over R; see also [MS05, Chapter 6]. One can visualize F as being supported on this simplicial complex below:





Its free modules are

$$\begin{aligned} F_0 &= R, \\ F_1 &= Re_1 + Re_2 + Re_3 + Re_4 + Re_5, \\ F_2 &= Re_{12} + Re_{13} + Re_{14} + Re_{23} + Re_{24} + Re_{34} + Re_{35} + Re_{45}, \\ F_3 &= Re_{123} + Re_{124} + Re_{134} + Re_{234} + Re_{345}, \\ F_4 &= Re_{1234}. \end{aligned} \tag{15}$$

For an ordered face $\sigma = (i_1 < \cdots < i_p)$, let m_σ be the least common multiple of its vertex labels. Our orientation convention is

$$\mathrm{d}e_\sigma = \sum_{j=1}^{p} (-1)^{j-1} \frac{m_\sigma}{m_{\sigma\setminus\{i_j\}}} e_{\sigma\setminus\{i_j\}}. \tag{16}$$

In particular, we have $\mathrm{d}e_1 = x^2$, $\mathrm{d}e_2 = w^2$, $\mathrm{d}e_3 = zw$, $\mathrm{d}e_4 = xy$, and $\mathrm{d}e_5 = y^2z^2$. Similarly we have

$$\begin{aligned} \mathrm{d}e_{123} &= x^2e_{23} - we_{13} + ze_{12}, \\ \mathrm{d}e_{124} &= xe_{24} - w^2e_{14} + ye_{12}, \\ \mathrm{d}e_{134} &= xe_{34} - zwe_{14} + ye_{13}, \\ \mathrm{d}e_{234} &= we_{34} - ze_{24} + xye_{23}, \\ \mathrm{d}e_{345} &= we_{45} - xe_{35} + yze_{34}, \end{aligned} \tag{17}$$

and

$$\mathrm{d}e_{1234} = -ye_{123} + ze_{124} - we_{134} + xe_{234}. \tag{18}$$

For completeness, exactness can also be seen without invoking the Scarf construction. If

$$J = \langle x^2, w^2, zw, xy \rangle, \qquad f = y^2z^2,$$

then $J : f = \langle x, w \rangle$. The Taylor complex on the first four generators is a minimal resolution of R/J over R, and the Koszul complex over R on x, w resolves $R/\langle x, w \rangle$. The mapping cone lifting

$$0 \longrightarrow R/\langle x, w \rangle \xrightarrow{f} R/J \longrightarrow R/I \longrightarrow 0$$

is the complex (15)–(18), with the indicated choices of signs. All differential entries belong to $\mathfrak{m}$, so this resolution is minimal. A more general version of this construction appears in Subsection 3.4.

3.2 The multiplication

In this subsection we construct a commutative multiplication on F which will serve as a reference. Let T be the Taylor resolution over R on all five generators, with its usual multiplication μ_T as recalled in [Kat19, Section 1.2], and let $\iota\colon F \hookrightarrow T$ be the inclusion. We specify a chain retraction $\pi\colon T \to F$. It is the identity on every basis face of F and has the following values on the remaining faces:

$$\begin{aligned}
\pi(e_{15}) &= yz^2e_{14} + xe_{45}, & \pi(e_{25}) &= y^2ze_{23} + we_{35},\\
\pi(e_{125}) &= yz^2e_{124} + xyze_{234} - xwe_{345}, & \pi(e_{135}) &= yze_{134} - xe_{345},\\
\pi(e_{145}) &= 0, & \pi(e_{235}) &= 0,\\
\pi(e_{245}) &= -yze_{234} + we_{345}, & \pi(e_{1235}) &= yze_{1234},\\
\pi(e_{1245}) &= 0, & \pi(e_{1345}) &= 0,\\
\pi(e_{2345}) &= 0, & \pi(e_{12345}) &= 0.
\end{aligned} \tag{19}$$

The Taylor differential directly gives $\mathrm{d}\pi = \pi\mathrm{d}$. Since $\pi\iota = \mathrm{id}_F$, the formula

$$\mu = \pi\mu_T\iota^{\otimes 2} \tag{20}$$

defines a commutative multiplication. This fixes the reference product on all pairs, not only on the pairs needed below. We write $ab = \mu(a,b)$. By Lemma 1.4, every competing multiplication has the form $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$, with h normalized. For a fixed such h, write $a \star b = \mu_h(a,b)$, reserving juxtaposition for the reference multiplication.

The subcomplex on the first four vertices is closed under μ, and its multiplication is exactly its Taylor multiplication. In particular, μ is associative on all triples from e_1, e_2, e_3, e_4. Its degree-one products on those generators are

$$e_1e_2 = e_{12}, \quad e_1e_3 = e_{13}, \quad e_1e_4 = xe_{14}, \quad e_2e_3 = we_{23}, \quad e_2e_4 = e_{24}, \quad e_3e_4 = e_{34}. \tag{21}$$

Other products used in the proof are

$$\begin{aligned}
e_1e_5 &= yz^2e_{14} + xe_{45}, & e_2e_5 &= y^2ze_{23} + we_{35},\\
e_1e_{23} &= e_{123}, & e_2e_{14} &= -e_{124},\\
e_1e_{35} &= yze_{134} - xe_{345}, & e_2e_{45} &= -yze_{234} + we_{345}.
\end{aligned} \tag{22}$$

They give

$$\begin{aligned}
[e_1, e_5, e_2]_\mu &= -yz^2e_{124} - xyze_{234} + xwe_{345} + y^2ze_{123} + wyze_{134} - wxe_{345}\\
&= -yz\,\mathrm{d}e_{1234}.
\end{aligned}$$

Finally, the products $e_iF_3 \subseteq F_4$ for $i \le 4$ are recorded in the following table; each entry denotes the coefficient of e_{1234}.

	e_{123}	e_{124}	e_{134}	e_{234}	e_{345}
e_1	0	0	0	x	0
e_2	0	0	$-w$	0	0
e_3	0	w	0	0	0
e_4	$-x$	0	0	0	0

(23)

Consequently,

$$e_iF_3 + F_3e_i \subseteq \langle x, w\rangle F_4 \qquad (1 \le i \le 4). \tag{24}$$

This property belongs to the fixed reference product. No corresponding property is assumed for an arbitrary multiplication μ_h. The computation above gives

$$[e_1, e_5, e_2]_\mu = -yz\, \mathrm{d}e_{1234} \in \mathfrak{m}^3 F_3. \tag{25}$$

Thus the nonmembership hypothesis of Theorem 1.1 fails for this triple and this reference multiplication.

Example 3.1. Define the normalized graded R-linear map $h\colon F \otimes_R F \to F$ of homological degree one by

$$h(e_1, e_5) = e_{345}, \qquad h(e_5, e_1) = -e_{345}, \qquad h(e_2, e_{34}) = e_{1234} = h(e_{34}, e_2), \tag{26}$$

and set it equal to zero on all other ordered pairs of basis elements. Then $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$ is a commutative multiplication, and the scalar variation formula in Lemma 3.4 gives

$$[e_1, e_5, e_2]_{\mu_h} = 0, \qquad [e_2, e_3, e_4]_{\mu_h} = -\mathrm{d}e_{1234} \neq 0. \tag{27}$$

Thus (e_1, e_5, e_2) cannot give a uniform non-associativity obstruction: the first associator can be cancelled, while another one becomes nonzero.

The obstruction will instead be relative to associativity on certain triples involving the first four degree-one basis elements. Their unit-coefficient products in (21) constrain the same homotopy coefficients which could cancel (25).

3.3 The relative obstruction

The proof uses associativity on triples from the first four generators to control the corrections to the associator at (e_1, e_5, e_2). We proceed in four steps.

1. Use $F_4 = Re_{1234}$ and injectivity of $\mathrm{d}\colon F_4 \to F_3$ to express each associator on three degree-one inputs as a coefficient times $\mathrm{d}e_{1234}$.
2. Localize at $\mathfrak{m}$ and reduce modulo $\langle x, w\rangle$. The reference products $e_i F_3 + F_3 e_i$ vanish for $i \leq 4$, while the reductions of the four products $e_1e_2, e_1e_3, e_2e_4, e_3e_4$ give a basis of the relevant submodule of the reduced F_2.
3. Impose associativity on the 32 triples specified below. Their equations and Nakayama's lemma force the relevant reduced homotopy maps to vanish on that submodule.
4. Evaluate the remaining associator at (e_1, e_5, e_2). Its coefficient is $-yz$ times a unit, so it cannot vanish.

Throughout this subsection, let $U = \{1, 4\}$, let $V = \{2, 3\}$, and define the set of 32 ordered triples

$$\mathcal{C} = (U \times U \times V) \cup (V \times V \times U) \cup (U \times V \times V) \cup (V \times U \times U). \tag{28}$$

Equivalently, $\mathcal{C}$ consists of the triples whose first and third indices lie in opposite sets U, V; the middle index is arbitrary in $\{1, 2, 3, 4\}$. Repeated indices are allowed.

Theorem 3.2. *The minimal free resolution F of R/I over R admits no associative multiplication. The same assertion holds for the corresponding minimal resolutions over $R_\mathfrak{m}$ and over $\Bbbk[\![x, y, z, w]\!]$. These assertions hold in every characteristic.*

We prove the following stronger statement.

Theorem 3.3 (Relative associativity obstruction)**.** *Let μ_h be any multiplication on F. If*

$$[e_i, e_j, e_k]_{\mu_h} = 0 \qquad \text{for every } (i,j,k) \in \mathcal{C},$$

then

$$[e_1, e_5, e_2]_{\mu_h} \neq 0.$$

In particular, among the 33 *triples in $\mathcal{C} \cup \{(1,5,2)\}$, at least one is non-associative for every multiplication.*

Theorem 3.2 follows immediately from Theorem 3.3. The relative theorem asserts that associativity on $\mathcal{C}$ prevents cancellation of (25).

3.3.1 Two lemmas on homotopy coefficients

The reduction to scalar associators uses the rank-one module F_4. All statements in this section apply after localization or completion as well.

By Lemma 1.4, write $\mu_h = \mu + \mathrm{d}h + h\mathrm{d}$ with h normalized. Since $F_{\geq 5} = 0$, its only potentially nonzero components on pairs of positive homological degrees are

$$h\colon F_1 \otimes_R F_1 \to F_3, \qquad h\colon F_1 \otimes_R F_2 \to F_4, \qquad h\colon F_2 \otimes_R F_1 \to F_4.$$

No symmetry of these components is imposed.

Fix such a normalized h and write

$$h_{ij} = h(e_i, e_j) \in F_3, \qquad h(e_i, a) = \ell_i(a) e_{1234}, \qquad h(a, e_i) = \ell_i'(a) e_{1234} \quad (a \in F_2), \tag{29}$$

where $\ell_i, \ell_i'\colon F_2 \to R$ are R-linear maps. Since $\mathrm{d}e_i, \mathrm{d}e_j \in R$ and h vanishes when either input is a scalar, the $h\mathrm{d}$ term vanishes on $e_i \otimes e_j$. Therefore

$$e_i \star e_j = e_i e_j + \mathrm{d}h_{ij}. \tag{30}$$

In particular, neither $h_{ii} = 0$ nor $h_{ji} = -h_{ij}$ is being assumed.

For any multiplication ν, the element $[e_i, e_j, e_k]_\nu$ is a cycle in F_3. Indeed, the Leibniz formula for the associator reduces its differential to associators with a scalar input, and those vanish by unitality and R-bilinearity. Exactness and injectivity of $\mathrm{d}\colon F_4 \to F_3$ therefore give unique $a_{ijk}^\nu \in R$ such that

$$[e_i, e_j, e_k]_\nu = a_{ijk}^\nu \, \mathrm{d}e_{1234}. \tag{31}$$

Lemma 3.4 (Scalar variation formula)**.** *For every ordered triple $i, j, k \in \{1, \dots, 5\}$, one has*

$$\big(a_{ijk}^{\mu_h} - a_{ijk}^{\mu}\big) e_{1234} = [e_i, e_j, e_k]_{\mu,h} + [e_i, e_j, e_k]_{h,\mu_h}.$$

Equivalently,

$$\begin{aligned} a_{ijk}^{\mu_h} e_{1234} = a_{ijk}^{\mu} e_{1234} + h_{ij} e_k + e_i h_{jk} \\ + \big(\ell_k'(e_i \star e_j) - \ell_i(e_j \star e_k)\big) e_{1234}. \end{aligned} \tag{32}$$

This identity does not require graded commutativity of μ_h.

Proof. Let $H = [\cdot]_{\mu,h} + [\cdot]_{h,\mu_h}$ as in (1). Normalization of h and unitality of the multiplications imply that the two generalized associators cancel whenever an input is a scalar. Thus H vanishes on such tensors. Since $\mathrm{d}e_i, \mathrm{d}e_j, \mathrm{d}e_k \in R$, we have $H\mathrm{d}(e_i \otimes e_j \otimes e_k) = 0$. Consequently, (1) identifies the difference of the two associators with $\mathrm{d}H(e_i, e_j, e_k)$. Together with (31) and the injectivity of $\mathrm{d}\colon F_4 \to F_3$, this proves the first identity. Since $|e_i| = 1$, evaluating the generalized associators gives

$$[e_i, e_j, e_k]_{\mu,h} = h_{ij}e_k + e_i h_{jk},$$
$$[e_i, e_j, e_k]_{h,\mu_h} = \big(\ell'_k(e_i \star e_j) - \ell_i(e_j \star e_k)\big)e_{1234}.$$

Substituting these expressions yields (32), the scalar expression of (1). □

The following lemma controls the relevant homotopy coefficients by Nakayama's lemma.

Lemma 3.5. *Let $(B, \mathfrak{n})$ be a commutative local ring, and let G be a free B-module with basis $e_{12}, e_{13}, e_{24}, e_{34}$. Use $U, V, \mathcal{C}$ from (28). Let $p_{ij} \in G$ for $i, j \in \{1,2,3,4\}$ be alternating, meaning $p_{ii} = 0$ and $p_{ji} = -p_{ij}$ for all i, j, with*

$$p_{12} = e_{12}, \quad p_{13} = e_{13}, \quad p_{24} = e_{24}, \quad p_{34} = e_{34}, \qquad p_{14} = p_{23} = 0.$$

Suppose $q_{ij} \in G$ satisfy $q_{ij} - p_{ij} \in \mathfrak{n}G$ for all $i, j \in \{1,2,3,4\}$. Note that no alternation is assumed for the q_{ij}. Let $\ell_i, \ell'_i \in \mathrm{Hom}_B(G, B)$ for $i \in \{1,2,3,4\}$. If

$$\ell'_k(q_{ij}) = \ell_i(q_{jk}) \qquad ((i,j,k) \in \mathcal{C}), \tag{33}$$

then all eight linear maps ℓ_i, ℓ'_i are zero.

Proof. Let $J \subseteq B$ be the ideal generated by the 32 values of the eight linear maps on the four displayed basis vectors, thus

$$J = \left\langle \begin{array}{l} \ell_i(e_{12}), \ell_i(e_{13}), \ell_i(e_{24}), \ell_i(e_{34}), \\ \ell'_i(e_{12}), \ell'_i(e_{13}), \ell'_i(e_{24}), \ell'_i(e_{34}) \end{array} \;\middle|\; 1 \le i \le 4 \right\rangle_B.$$

We show $J \subseteq \mathfrak{n}J$.

Fix $i \in U$. Let j range through U and k through V. Then $p_{ij} = 0$, so $q_{ij} \in \mathfrak{n}G$, while the four elements p_{jk} run through a basis of G up to signs. From (33),

$$\ell_i(p_{jk}) = \ell'_k(q_{ij}) - \ell_i(q_{jk} - p_{jk}) \in \mathfrak{n}J.$$

Thus $\ell_i(G) \subseteq \mathfrak{n}J$. The same argument with U, V interchanged applies to $i \in V$.

Next fix $k \in U$, let j range through U, and let i range through V. Now $q_{jk} \in \mathfrak{n}G$, and the four p_{ij} again form a basis up to signs. Thus

$$\ell'_k(p_{ij}) = \ell_i(q_{jk}) - \ell'_k(q_{ij} - p_{ij}) \in \mathfrak{n}J.$$

Interchanging U, V gives the assertion for $k \in V$. Every generator of J belongs to $\mathfrak{n}J$. Hence $J = \mathfrak{n}J$, and Nakayama's lemma gives $J = 0$. □

3.3.2 Proof of the relative obstruction

Proof of Theorem 3.3. It suffices to work over $R_{\mathfrak{m}}$: any multiplication over R localizes to one over $R_{\mathfrak{m}}$. Put

$$B = R_{\mathfrak{m}}/\langle x, w\rangle \cong \Bbbk[y,z]_{\langle y,z\rangle}, \qquad \mathfrak{n} = \langle y, z\rangle B,$$

and use a bar for reduction modulo $\langle x, w\rangle$. Inside $\overline{F}_2 = F_2 \otimes B$, let

$$G = B\bar{e}_{12} + B\bar{e}_{13} + B\bar{e}_{24} + B\bar{e}_{34}. \tag{34}$$

Equations (17) give

$\bar{a}$	$\bar{e}_{123}$	$\bar{e}_{124}$	$\bar{e}_{134}$	$\bar{e}_{234}$	$\bar{e}_{345}$
$\mathrm{d}\bar{a}$	$z\bar{e}_{12}$	$y\bar{e}_{12}$	$y\bar{e}_{13}$	$-z\bar{e}_{24}$	$yz\bar{e}_{34}.$

(35)

Consequently $\mathrm{d}\overline{F}_3 \subseteq \mathfrak{n}G$. Put $p_{ij} = \overline{e_i e_j}$ and $q_{ij} = \overline{e_i \star e_j}$ for $i, j \le 4$. Thus p_{ij} is the reduction of the reference product, whereas q_{ij} is the reduced perturbed product. For $i, j \le 4$, (21) and (30) show

$$q_{ij} \in G, \qquad q_{ij} - p_{ij} \in \mathfrak{n}G. \tag{36}$$

The reference products satisfy $p_{14} = p_{23} = 0$, and $p_{12}, p_{13}, p_{24}, p_{34}$ form the displayed basis of G. These are the elements in Lemma 3.5, with its abstract basis elements $e_{12}, e_{13}, e_{24}, e_{34}$ identified with $\bar{e}_{12}, \bar{e}_{13}, \bar{e}_{24}, \bar{e}_{34}$, respectively.

The hypothesis of the theorem says $a_{ijk}^{\mu_h} = 0$ for $(i,j,k) \in \mathcal{C}$. The reference product is associative on the first four generators, so $a_{ijk}^{\mu} = 0$ there as well. The two μ-terms in (32) vanish after reduction by (24). We therefore obtain

$$\bar{\ell}'_k(q_{ij}) = \bar{\ell}_i(q_{jk}) \qquad ((i,j,k) \in \mathcal{C}).$$

Apply Lemma 3.5 with $q_{ij} = \overline{e_i \star e_j}$ and the B-linear maps $\bar{\ell}_i|_G, \bar{\ell}'_i|_G$. It gives

$$\bar{\ell}_i(G) = 0 = \bar{\ell}'_i(G) \qquad (1 \le i \le 4). \tag{37}$$

Now apply (32) to $(i,j,k) = (1,5,2)$. By (24), the reference-product terms $h_{15}e_2 + e_1 h_{52}$ vanish modulo $\langle x, w\rangle$. Equations (30) and (22) give

$$\overline{e_1 \star e_5} = yz^2\bar{e}_{14} + \mathrm{d}\bar{h}_{15}, \qquad \overline{e_5 \star e_2} = -y^2 z\bar{e}_{23} + \mathrm{d}\bar{h}_{52}.$$

Since $h_{15}, h_{52} \in F_3$, the differential table (35) places both $\mathrm{d}\bar{h}_{15}$ and $\mathrm{d}\bar{h}_{52}$ in $\mathrm{d}\overline{F}_3 \subseteq \mathfrak{n}G$. Thus (37) gives $\bar{\ell}'_2(\mathrm{d}\bar{h}_{15}) = 0$ and $\bar{\ell}_1(\mathrm{d}\bar{h}_{52}) = 0$. Using $a_{152}^{\mu} = -yz$, the scalar variation formula becomes

$$\begin{aligned} \bar{a}_{152}^{\mu_h} &= -yz + \bar{\ell}'_2(\overline{e_1 \star e_5}) - \bar{\ell}_1(\overline{e_5 \star e_2}) \\ &= -yz + yz^2\bar{\ell}'_2(\bar{e}_{14}) + y^2 z\bar{\ell}_1(\bar{e}_{23}) \\ &= -yz\big(1 - z\bar{\ell}'_2(\bar{e}_{14}) - y\bar{\ell}_1(\bar{e}_{23})\big). \end{aligned} \tag{38}$$

The parenthesized factor is a unit of B, since its residue in $B/\mathfrak{n}$ is 1. The ring B is a domain and $yz \neq 0$, so (38) is nonzero. Hence $a_{152}^{\mu_h} \neq 0$, and the injectivity of $\mathrm{d}\colon F_4 \to F_3$ gives $[e_1, e_5, e_2]_{\mu_h} \neq 0$.

For the completed ring, the same proof uses $B = \Bbbk[\![y,z]\!]$. This is again a local domain, and every step is unchanged. □

Remark 3.6. Exactness is used before reduction modulo $\langle x, w\rangle$, to normalize the homotopy and to identify scalar associators. The reduced complex is not assumed to remain a resolution over B. After reduction, the proof uses only the displayed matrices, linear maps, and Nakayama's lemma. In particular, no exactness assertion is being transported through a non-flat quotient.

Likewise, no step assumes $e_i \star e_i = 0$ or $e_j \star e_i = -e_i \star e_j$. Alternation occurs only in the fixed leading products p_{ij} of the reference multiplication. The perturbations $\overline{e_i \star e_j} - p_{ij}$ and the left and right linear maps remain unrestricted.

3.4 A family with higher-order reference obstructions

The argument is not specific to the coefficient yz in (25). Let

$$0 \neq c(y,z) \in \langle y,z\rangle \Bbbk[y,z], \qquad I_c = \langle x^2, w^2, zw, xy, yz\,c(y,z)\rangle.$$

The polynomial c need not be homogeneous or a monomial. Put $A = R_{\mathfrak{m}}$, with $R = \Bbbk[x,y,z,w]$. We state minimality over A; the construction also provides a polynomial free resolution over R.

Theorem 3.7. *The minimal free resolution of A/I_cA over A admits no associative multiplication. More precisely, associativity on the 32 core triples forces non-associativity at (e_1, e_5, e_2). The same assertions hold for the explicit polynomial free resolution over R constructed below and after completion of A at its maximal ideal.*

Proof. Let $J = \langle x^2, w^2, zw, xy\rangle$ and $f = yzc$. We first verify $J : f = \langle x, w\rangle$. Certainly xf and wf belong to J. Conversely, if $gf \in J$, then reduction modulo $\langle x, w\rangle$ gives $\bar{g}\,yzc = 0$ in the domain $\Bbbk[y,z]$. Since $c \neq 0$, one has $\bar{g} = 0$, so $g \in \langle x, w\rangle$.

The mapping cone of $R/\langle x,w\rangle \xrightarrow{f} R/J$ has the same basis as (15). The core differentials are unchanged, and the remaining differentials are

$$\mathrm{d}e_5 = yzc, \qquad \mathrm{d}e_{35} = we_5 - yce_3, \qquad \mathrm{d}e_{45} = xe_5 - zce_4, \qquad \mathrm{d}e_{345} = we_{45} - xe_{35} + ce_{34}. \tag{39}$$

To see the comparison map explicitly, let a, b be the Koszul generators with $\mathrm{d}a = x$ and $\mathrm{d}b = w$. A lift to the Taylor complex on J is

$$1 \longmapsto yzc, \qquad a \longmapsto zce_4, \qquad b \longmapsto yce_3, \qquad ab \longmapsto -ce_{34}.$$

The usual cone differential, after choosing signs for its shifted basis, gives (39). Since $c \in \langle y,z\rangle$, every matrix entry is in $\mathfrak{m}$, and the localized cone is minimal. If c is homogeneous, the polynomial cone is the minimal graded resolution over R as well.

To construct a commutative reference multiplication, first work over $\Bbbk[x,y,z,w,t]$ with ideal $\langle x^2, w^2, zw, xy, yzt\rangle$. In (19), replace each occurrence of yz by t, each occurrence of y^2z by yt, and each occurrence of yz^2 by zt. These formulas define a Taylor chain retraction and hence a commutative multiplication. Substitute $t = c(y,z)$. Chain-map and product identities are polynomial identities and therefore survive this substitution; exactness of the specialized complex was established separately by the cone construction.

The reference product is still associative on the first four generators, its top table is still (23), and

$$e_1e_5 = zce_{14} + xe_{45}, \qquad e_2e_5 = yce_{23} + we_{35}, \qquad [e_1, e_5, e_2]_\mu = -c\,\mathrm{d}e_{1234}.$$

Modulo $\langle x,w\rangle$, the last entry of (35) becomes $c\bar{e}_{34}$. Since $c \in \mathfrak{n}$, the image of the reduced differential is still contained in $\mathfrak{n}G$. The Nakayama argument is unchanged. Assuming associativity on $\mathcal{C}$, the final scalar associator is

$$\bar{a}^{\mu_h}_{152} = -c\big(1 - z\bar{\ell}'_2(\bar{e}_{14}) - y\bar{\ell}_1(\bar{e}_{23})\big) \neq 0.$$

The factor in parentheses is a unit and c is nonzero in $\Bbbk[y,z]_{\langle y,z\rangle}$, or in $\Bbbk[\![y,z]\!]$ for the completed case. Localization and completion preserve exactness by flatness. Every polynomial multiplication localizes, so the same relative assertion holds on the displayed polynomial complex. This proves the theorem. □

Corollary 3.8. *For integers $a, b \geq 1$ with $a + b \geq 3$, put*

$$I_{a,b} = \langle x^2, w^2, zw, xy, y^a z^b \rangle.$$

The quotient $R/I_{a,b}$ has a minimal free resolution over R with Betti numbers $(1, 5, 8, 5, 1)$ which admits no associative multiplication. For the reference product constructed above,

$$[e_1, e_5, e_2]_\mu = -y^{a-1} z^{b-1} \mathrm{d}e_{1234}.$$

Proof. Apply Theorem 3.7 with $c = y^{a-1} z^{b-1}$. □

For example, when $a = b = r + 1$ with $r \geq 1$, the displayed associator has coefficients of order $2r + 1$ in $\mathfrak{m}$. Thus the reference obstruction can lie arbitrarily deep in the maximal ideal while nonexistence of an associative multiplication remains valid. The case $a = b = 1$ is deliberately excluded from the stated complex: then the coefficient of e_{34} in $\mathrm{d}e_{345}$ is a unit and this cone is not minimal. Cancelling that pair gives the smaller cellular resolution appearing in Avramov's example, studied in Subsection 1.3.

4 Obstructions to multiplicative Taylor comparisons

Throughout this section, let $\Bbbk$ be a field, let $R = \Bbbk[x_1, \ldots, x_n]$, let $\mathfrak{m} = \langle x_1, \ldots, x_n \rangle$, let $I \subsetneq R$ be a nonzero monomial ideal, and let F be the minimal multigraded free resolution of R/I over R, with $F_0 = R$. The resolution carries its fine multigrading, but competing multiplications and comparison maps need not preserve it.

Let $J \subseteq I$ be generated by a selection of minimal generators of I, and equip its Taylor resolution $T(J)$ with the usual associative multiplication μ_T [Kat19, Section 1.2]. A unital chain comparison $\varphi \colon T(J) \to F$ is an R-linear chain map with $\varphi_0 = \mathrm{id}_R$, lifting $R/J \twoheadrightarrow R/I$. It is multiplicative for a multiplication μ_h on F if

$$\varphi(ab) = \mu_h(\varphi(a), \varphi(b)) \qquad (a, b \in T(J)).$$

We ask how large a selection can admit such a comparison when the multiplication on F is allowed to vary.

Uniform non-associativity gives an obstruction that survives changes of both the multiplication and the degree-one lifts defining the comparison. For the edge ideal of the eight-cycle, the largest possible selection has four generators. Its big height is five, and an injective comparison on five selected generators exists, but every five-generator selection fails to admit a multiplicative comparison. We also obtain a family of cycle edge ideals with this failure at the big-height size.

For squarefree ideals, Katthän realizes a multigraded DG algebra resolution as a quotient of the full Taylor algebra [Kat19, Theorem 3.6]. VandeBogert extends this to generalized Taylor resolutions [VdB23, Section 4], and Ferraro and Alvarez give a divided-power version [FA25, Theorem 5.4]. Those results assume an associative multigraded target. Our question concerns Taylor algebras on selected generators and permits a nonassociative target.

This question is motivated by multiplicative approaches to rank bounds. Buchsbaum and Eisenbud use an associative commutative multiplication on a resolution to obtain an injective comparison from the Koszul algebra on a regular sequence [BE77, Proposition 1.4]. Here the source is a Taylor algebra on selected minimal generators, and the target multiplication is allowed to be nonassociative.

4.1 Injective Taylor comparisons and rank bounds

We first record an injectivity consequence of the associated-prime characterization by dominant sets in [BEK+25, Theorem 2.1]. That result supplies the selected generators and the pure-power ideal used below; we include the construction to make the comparison-map argument explicit. The proposition distinguishes failure of multiplicativity from a failure of the underlying numerical bound.

Proposition 4.1 (Injective Taylor comparisons). *Let $I \subsetneq R$ be a nonzero monomial ideal, and let*

$$m = \operatorname{bight} I := \max\{\operatorname{ht} \mathfrak{p} \mid \mathfrak{p} \in \operatorname{Ass}_R(R/I)\}.$$

There are m distinct minimal generators $u_1, \dots, u_m$ of I such that R/J, where $J = \langle u_1, \dots, u_m \rangle$, has a minimal Taylor resolution T over R, and every unital multigraded comparison $\varphi\colon T \to F$ lifting $R/J \twoheadrightarrow R/I$ is injective. In particular,

$$\beta_i^R(R/I) \geq \binom{m}{i}.$$

Proof. Choose $\mathfrak{p} \in \operatorname{Ass}_R(R/I)$ with $\operatorname{ht} \mathfrak{p} = m$. Since I is monomial, we can write $\mathfrak{p} = \langle z_1, \dots, z_m \rangle$, where $z_j = x_{i_j}$ for distinct indices $i_1, \dots, i_m \in \{1, \dots, n\}$. Thus $z_1, \dots, z_m$ is an ordered selection of the original variables, and $m \leq n$. Choose a monomial $v \notin I$ such that $I : v = \mathfrak{p}$, and put $a_j = \deg_{z_j}(v) + 1$. Since $z_j v \in I$, choose a minimal generator u_j of I dividing $z_j v$. Since $u_j \nmid v$, we have

$$\deg_{z_j}(u_j) = a_j, \qquad \deg_{z_{j'}}(u_j) < a_{j'} \quad (j' \neq j). \tag{40}$$

The selected generators are distinct. For each j, the z_j-exponent of u_j is strictly larger than that of every other selected generator. Consequently every coefficient in the positive Taylor differential belongs to $\mathfrak{m}$. Hence the Taylor resolution of R/J over R is minimal. The selected generators form a dominant set in the terminology of [Ale17, Definition 4.1], whose Theorem 4.4 characterizes minimal Taylor resolutions by this condition.

Set $Q = \langle z_1^{a_1}, \dots, z_m^{a_m} \rangle$. We claim that $I \subseteq Q$. For a minimal generator u of I, the monomial $r = u/\gcd(u, v)$ belongs to $I : v = \mathfrak{p}$. Thus some z_j divides r, which means $\deg_{z_j}(u) \geq a_j$. This proves the claim.

Let $E = K_R(z_1^{a_1}, \dots, z_m^{a_m})$, and choose a unital multigraded comparison $\psi\colon F \to E$ lifting $R/I \twoheadrightarrow R/Q$. The inclusions $J \subseteq I \subseteq Q$ give chain comparisons

$$T \xrightarrow{\varphi} F \xrightarrow{\psi} E.$$

We show that their composition is injective. For $A \subseteq \{1, \dots, m\}$, put

$$u_A = \operatorname{lcm}(u_j : j \in A), \qquad q_A = \prod_{j \in A} z_j^{a_j}.$$

The Taylor algebra T and the Koszul algebra E are both supported on the full simplex on $\{1, \dots, m\}$, with respective face labels u_A and q_A. Write t_A and ε_A for the Taylor and Koszul basis elements, respectively. A coefficient from t_A to ε_B, with $|A| = |B|$, can be nonzero only if $q_B \mid u_A$. By (40), this forces $B \subseteq A$ and hence $B = A$. Thus every unital multigraded chain map $\theta\colon T \to E$ is diagonal. Commuting with the differentials, starting with $\theta_0 = 1$, gives

$$\theta(t_A) = \frac{u_A}{q_A} \varepsilon_A.$$

All diagonal coefficients are nonzero monomials. Taking $\theta = \psi\varphi$, we see that $\psi\varphi$ is injective, and hence φ is injective. The rank inequality follows. □

For Katthän's ideal $I = \langle x^2, w^2, zw, xy, y^2z^2 \rangle$, take $v = xw$. Then $v \notin I$ and $I : v = \langle x, w, z, y \rangle = \mathfrak{m}$, so $m = n = 4$. The construction gives

$$(z_1, z_2, z_3, z_4) = (x, w, z, y), \qquad (a_1, a_2, a_3, a_4) = (2, 2, 1, 1),$$
$$(u_1, u_2, u_3, u_4) = (x^2, w^2, zw, xy), \qquad Q = \langle x^2, w^2, z, y \rangle.$$

The exponents are strictly largest among these four selected generators; the omitted generator y^2z^2 has larger y- and z-exponents. In degree one, the diagonal comparison sends (t_1, t_2, t_3, t_4) to $(\varepsilon_1, \varepsilon_2, w\varepsilon_3, x\varepsilon_4)$. For the reference multiplication in Subsection 3.2, this four-generator Taylor algebra is a subalgebra of F.

For comparison, put $c = \operatorname{ht} I$ and $\rho = \operatorname{pd}_R(R/I)$. Then

$$c \leq m = \operatorname{bight} I \leq \rho.$$

The last inequality follows from the Auslander–Buchsbaum formula after localization at an associated prime. Brun and Römer prove the stronger numerical bound

$$\beta_i^R(R/I) \geq \binom{\rho}{i}$$

for monomial quotients over every field [BR04, Corollary 4.1(i)]. For related binomial bounds from dominant sets, see [Ale23, Theorem 3.5]. Proposition 4.1 supplies specific injective comparisons to which we can apply the question of multiplicativity.

4.2 Changing lifts and the induced-path obstruction

The obstruction must be independent of the choice of comparison. The following elementary consequence of minimality is useful for this purpose.

Lemma 4.2 (Changing the degree-one lifts)**.** *Let $(R, \mathfrak{m})$ be local and let F be a minimal free resolution of R/I over R with $F_0 = R$, and let μ be a multiplication on F. Then*

$$[F, F, F]_\mu \subseteq \mathfrak{m}F.$$

If $a_j, a_j' \in F_1$ and $a_j' - a_j \in \mathfrak{m}F_1$ for $1 \leq j \leq 3$, then

$$[a_1', a_2', a_3']_\mu \equiv [a_1, a_2, a_3]_\mu \pmod{\mathfrak{m}^2F_3}. \tag{41}$$

In particular, an associator which is nonzero modulo $\mathfrak{m}^2F_3$ cannot be removed by changing these degree-one lifts.

Proof. Put $\Bbbk = R/\mathfrak{m}$ and $F_\Bbbk = F \otimes_R \Bbbk$. By minimality, the differential on $F_\Bbbk$ vanishes, so

$$F_\Bbbk = H(F_\Bbbk) \cong \operatorname{Tor}^R(R/I, \Bbbk).$$

The product induced by μ is the usual associative product on this Tor algebra. Indeed, the two iterated products $\mu(\mu \otimes 1)$ and $\mu(1 \otimes \mu)$ on F lift the same triple product on R/I, so they are chain homotopic. Tensoring this homotopy with $\Bbbk$ makes the two maps equal because the differentials vanish. Thus tensoring with $\Bbbk$ kills every associator, giving $[F, F, F]_\mu \subseteq \mathfrak{m}F$.

For $z \in \mathfrak{m}$ and $a, b, c \in F_1$, trilinearity and the first assertion give

$$[za, b, c]_\mu = z[a, b, c]_\mu \in \mathfrak{m}^2F_3.$$

Taking finite sums gives the same conclusion whenever the first input lies in $\mathfrak{m}F_1$, and the same argument applies in the other two positions. Since $a'_j - a_j \in \mathfrak{m}F_1$, replacing one input at a time yields

$$\begin{aligned}[a'_1, a'_2, a'_3]_\mu &\equiv [a_1, a'_2, a'_3]_\mu \\ &\equiv [a_1, a_2, a'_3]_\mu \\ &\equiv [a_1, a_2, a_3]_\mu \pmod{\mathfrak{m}^2 F_3}.\end{aligned}$$

This proves (41). □

Suppose now that T is a Taylor resolution over R on a selection of minimal generators of I, and $t_i \in T_1$ and $e_i \in F_1$ have the same differential u_i. For every unital chain comparison $\varphi\colon T \to F$,

$$\varphi(t_i) - e_i \in \ker(\mathrm{d}\colon F_1 \to F_0) = \mathrm{d}F_2 \subseteq \mathfrak{m}F_1. \tag{42}$$

Thus Lemma 4.2 applies to the images under every comparison, not only to a fixed multigraded comparison.

Proposition 4.3. *Let Γ be a finite simple graph, let $I(\Gamma)$ be its edge ideal, and let F be the minimal multigraded free resolution of $R/I(\Gamma)$ over R. Suppose Γ contains an induced path*

$$v_1 - v_2 - v_3 - v_4 - v_5 - v_6.$$

Let J be generated by a selection of edges of Γ which includes

$$v_1v_2, \qquad v_3v_4, \qquad v_5v_6,$$

and give $T(J)$ its usual Taylor multiplication. Then no unital chain comparison $T(J) \to F$ lifting $R/J \twoheadrightarrow R/I(\Gamma)$ is multiplicative for any multiplication on F. Neither the comparison nor the competing multiplication is required to respect an internal grading.

Proof. The path calculation in Subsection 2.2 satisfies the hypotheses of Theorem 1.1. Since the path is induced, its edge ideal is an elimination ideal of $I(\Gamma)$. Proposition 2.2 transfers those hypotheses to F at the degree-one elements a_1, a_2, a_3 corresponding to the three displayed alternating edges.

The proof of the uniform criterion gives more than nonvanishing: for every multiplication ν,

$$[a_1, a_2, a_3]_\nu \notin \mathfrak{m}^2 F_3.$$

Indeed, if this associator belonged to $\mathfrak{m}^2F_3$, the variation congruence (3) would place the reference associator in the submodule excluded by the criterion. The same reasoning applies after the elimination step. One may localize at the homogeneous maximal ideal throughout.

Let $t_1, t_2, t_3 \in T(J)_1$ denote the corresponding Taylor generators. Equations (42) and (41) imply

$$[\varphi(t_1), \varphi(t_2), \varphi(t_3)]_\nu \neq 0.$$

But multiplicativity and associativity of the Taylor algebra would give

$$[\varphi(t_1), \varphi(t_2), \varphi(t_3)]_\nu = \varphi([t_1, t_2, t_3]_{\mu_T}) = 0,$$

a contradiction. □

The Taylor resolution in Proposition 4.3 need not be minimal. The obstruction rules out multiplicativity itself, before any question about injectivity is considered.

4.3 Cycle edge ideals

The cycle edge ideal considered here is different from the ideal of the eight-vertex Bier sphere in Theorem 2.4. Its role is to obstruct every sufficiently large Taylor selection.

Theorem 4.4. *Let* $R = \Bbbk[x_1, \dots, x_8]$ *and*

$$I = \langle u_1, \dots, u_8 \rangle, \qquad u_i = x_i x_{i+1},$$

where indices are read modulo 8. *Let* F *be the minimal free resolution of* R/I *over* R. *Then* $\operatorname{ht} I = 4$ *and* $\operatorname{bight} I = 5$. *For every five-element subset* $S \subseteq \{1, \dots, 8\}$, *put* $J_S = \langle u_i : i \in S \rangle$ *and let* T_S *be its Taylor algebra. No unital comparison* $T_S \to F$ *is multiplicative for any multiplication on* F.

Nevertheless, a five-generator selection has a minimal Taylor resolution over R *admitting a degreewise injective comparison into* F, *and a four-generator selection admits a multiplicative embedding for a suitable multigraded multiplication on* F. *Thus the largest number of selected minimal generators whose Taylor algebra can map multiplicatively into* F *is exactly four. These assertions hold over every field.*

Proof. The four disjoint edges $12, 34, 56, 78$ show that every vertex cover has at least four vertices, while the even vertices form a cover of size four. Thus $\operatorname{ht} I = 4$.

The complements of maximal independent sets are the minimal vertex covers. Every maximal independent set of C_8 is dominating, and a vertex together with its neighbors contains at most three vertices. Hence every maximal independent set has at least three vertices. The set $\{2, 5, 7\}$ is a maximal independent set of size three. The largest minimal vertex cover therefore has size five, giving $\operatorname{bight} I = 5$.

Now let S have five elements. At least three elements of S have the same parity. Every three-element subset of either $\{1, 3, 5, 7\}$ or $\{2, 4, 6, 8\}$ has the form

$$\{i, i+2, i+4\} \pmod 8.$$

The corresponding generators are the alternating edges of the induced path on the six consecutive vertices $i, i+1, \dots, i+5$. Proposition 4.3 proves the nonexistence assertion.

To see that this is not a failure of injectivity, take

$$\begin{aligned} J_5 &= \langle u_1, u_2, u_4, u_5, u_7 \rangle \\ &= \langle x_1x_2, x_2x_3, x_4x_5, x_5x_6, x_7x_8 \rangle. \end{aligned}$$

Its generators have private variables x_1, x_3, x_4, x_6, x_8, respectively, so its Taylor resolution is minimal. Moreover,

$$I : \langle x_2x_5x_7 \rangle = \langle x_1, x_3, x_4, x_6, x_8 \rangle.$$

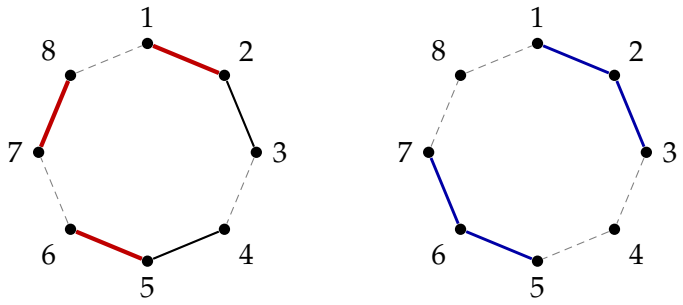
These are precisely the witness and selection used in Proposition 4.1. Thus the multigraded comparison $T(J_5) \to F$ is injective, although no comparison from this Taylor algebra can be multiplicative.

Finally, let

$$\begin{aligned} J_4 &= \langle u_1, u_2, u_5, u_6 \rangle \\ &= \langle x_1x_2, x_2x_3, x_5x_6, x_6x_7 \rangle. \end{aligned}$$

Put $R_0 = \Bbbk[x_1, x_2, x_3, x_5, x_6, x_7]$. Then $J_4 = (I \cap R_0)R$. The Taylor resolution of $R_0/(I \cap R_0)$ over R_0 is minimal, with private variables x_1, x_3, x_5, x_7, and its scalar extension is the Taylor resolution $T(J_4)$ of R/J_4 over R. Lemma 2.1 identifies this scalar extension with a basis subcomplex of F and extends its Taylor multiplication to a multigraded multiplication on F. The resulting inclusion is

multiplicative and injective. The same positive conclusion also follows from [Kat19, Proposition 3.1]: the four selected generators span a full Scarf simplex, since they have private variables and every omitted edge uses x_4 or x_8. A multiplicative comparison on more than five selected generators would restrict to one on five, which has already been excluded. This proves the exact maximum. □



$J_5 = \langle u_1, u_2, u_4, u_5, u_7 \rangle$ $J_4 = \langle u_1, u_2, u_5, u_6 \rangle$

Figure 1: Solid edges are the selected generators. In J_5, the three heavier edges u_5, u_7, u_1 alternate along the induced path $5-6-7-8-1-2$ and obstruct multiplicativity. Deleting vertices $4, 8$ leaves exactly J_4, whose Taylor algebra embeds multiplicatively.

Remark 4.5. The nonexistence of a commutative DG algebra structure on the minimal resolution of the eight-cycle is already a consequence of the classification of DG cycles in [GMP26, Theorem 6.2]. Theorem 4.4 concerns a different question. The target is allowed to remain nonassociative, and the multiplication, selection, and comparison may all vary. Even with this freedom, multiplicativity cannot be arranged on five selected generators. This contrasts with the four-generator Taylor subalgebra in Katthän's example.

Corollary 4.6. *For $r \geq 1$, put $R = \Bbbk[x_1, \ldots, x_{6r+2}]$, let I be the edge ideal of C_{6r+2} and let F be the minimal resolution of R/I over R. Then*

$$\operatorname{ht} I = 3r + 1, \qquad \operatorname{bight} I = 4r + 1.$$

For any selection of $4r+1$ edge generators, let J be the ideal they generate. No unital chain comparison $T(J) \to F$ lifting $R/J \twoheadrightarrow R/I$ is multiplicative for any multiplication on F. Neither the comparison nor the multiplication is required to respect the internal grading.

Proof. A perfect matching and a parity vertex cover give height $3r+1$. Every maximal independent set has at least $\lceil (6r+2)/3 \rceil = 2r+1$ vertices, and the set $\{1, 4, 7, \ldots, 6r+1\}$ attains this bound. Thus big height is $4r+1$.

The odd and even edge indices each form a cyclically ordered set of size $3r+1$ when successive indices differ by two. If a selection avoids all triples $\{i, i+2, i+4\}$, its indicator sequence on either parity class has no three consecutive ones. Summing the $3r+1$ cyclic inequalities

$$a_j + a_{j+1} + a_{j+2} \leq 2$$

gives $3\sum_j a_j \leq 2(3r+1)$, hence $\sum_j a_j \leq 2r$. Such a selection has at most $4r$ edges. Every selection of $4r+1$ edges therefore contains the alternating edges of an induced six-vertex path, and Proposition 4.3 applies. □

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the author used OpenAI Codex to assist with exposition, literature searches, exploration and checking of proof arguments, and the generation of symbolic-verification scripts. The author reviewed and revised the resulting material and takes responsibility for the mathematical claims, computations, and references.

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